

#1 · Acute Care and Diagnosis

What are the four Centor criteria used to estimate the probability of strep pharyngitis?

Answer

Fever by history, tonsillar exudate, tender anterior cervical adenopathy, and absence of cough. Each present feature scores one point. The modified McIsaac version adds one point for age under 15 and subtracts one point for age 45 or older.

#2 · Acute Care and Diagnosis

A patient has a Centor score of 1. What is the next step?

Answer

No testing and no antibiotics are needed. A score of 0 or 1 means strep is unlikely and the sore throat is almost certainly viral. Symptomatic treatment is appropriate.

#3 · Acute Care and Diagnosis

A patient has a Centor score of 3. What is the next step?

Answer

Order a rapid antigen detection test or throat culture. Antibiotics should only be given if the test result is positive. Testing before treating is the safer and more defensible approach.

#4 · Acute Care and Diagnosis

Why is cough considered a clue against strep pharyngitis?

Answer

Group A Streptococcus pharyngitis classically spares the lower respiratory tract. A prominent cough points toward a viral cause instead, which is why absence of cough is one of the Centor criteria that favors strep.

#5 · Acute Care and Diagnosis

What is the first line antibiotic for confirmed group A strep pharyngitis?

Answer

Penicillin V or amoxicillin for ten days.
Cephalosporins or macrolides are used in patients with a penicillin allergy.

#6 · Acute Care and Diagnosis

A patient with sore throat has drooling, a muffled hot potato voice, trismus, and uvular deviation. What should you suspect?

Answer

Peritonsillar abscess. This needs urgent drainage rather than routine oral antibiotics alone, so the patient should be referred for incision and drainage or aspiration.

#7 · Acute Care and Diagnosis

What is double sickening and why does it matter?

Answer

Double sickening is when a patient with a viral illness starts improving and then worsens again several days later. It is a classic sign of secondary bacterial infection, most often tested in the context of sinusitis but relevant across several respiratory illnesses.

#8 · Acute Care and Diagnosis

What three symptom patterns suggest bacterial rather than viral acute sinusitis?

Answer

Symptoms lasting ten days or more without improvement, severe symptoms with high fever and purulent discharge for at least three to four consecutive days at onset, or double sickening after initial improvement around day five or six.

#9 · Acute Care and Diagnosis

Does the color of nasal discharge reliably distinguish viral from bacterial sinusitis?

Answer

No. Purulent or colored discharge can occur in both viral and bacterial sinusitis, so discharge color alone should not be used to diagnose bacterial infection or justify antibiotics.

#10 · Acute Care and Diagnosis

What is the first line antibiotic for bacterial acute sinusitis?

Answer

Amoxicillin-clavulanate. It is preferred over amoxicillin alone because it better covers beta-lactamase producing organisms.

#11 · Acute Care and Diagnosis

A patient with sinusitis develops periorbital swelling, vision changes, and severe headache. What should you do?

Answer

Suspect orbital or intracranial extension of the sinus infection. This requires urgent CT imaging and urgent referral rather than routine outpatient antibiotic management.

#12 · Acute Care and Diagnosis

What three findings are required to diagnose acute otitis media?

Answer

Acute onset of symptoms, a middle ear effusion confirmed on otoscopy such as a bulging tympanic membrane or air-fluid level, and signs of middle ear inflammation such as marked erythema or new onset otalgia.

#13 · Acute Care and Diagnosis

Is a red tympanic membrane alone enough to diagnose acute otitis media?

Answer

No. Erythema without a confirmed middle ear effusion is not sufficient for the diagnosis. Crying alone can redden the eardrum, so effusion must be documented.

#14 · Acute Care and Diagnosis

In which age groups and severity levels is watchful waiting appropriate for acute otitis media?

Answer

Children 6 months to 2 years with unilateral mild disease and no perforation, and children 2 years or older with mild bilateral or unilateral disease, as long as follow-up in 48 to 72 hours is assured.

#15 · Acute Care and Diagnosis

When should amoxicillin-clavulanate be used instead of amoxicillin for acute otitis media?

Answer

When the child received amoxicillin in the past 30 days, has concurrent conjunctivitis, or has a history of recurrent acute otitis media that did not respond to amoxicillin.

#16 · Acute Care and Diagnosis

What exam finding is classic for otitis externa but not otitis media?

Answer

Pain with tragal traction or pinna manipulation. Otitis externa is an infection of the ear canal skin, so manipulating the outer ear reproduces the pain, unlike otitis media.

#17 · Acute Care and Diagnosis

What is the first line treatment for uncomplicated otitis externa?

Answer

Topical drops combining an antibiotic and a corticosteroid, such as ciprofloxacin-dexamethasone. Oral antibiotics are not first line because this is a localized skin infection of the canal.

#18 · Acute Care and Diagnosis

A diabetic patient has severe ear pain, granulation tissue in the canal, and possible cranial nerve findings. What is the diagnosis and next step?

Answer

Malignant, or necrotizing, otitis externa. This requires urgent imaging and IV antibiotics rather than outpatient topical drops, since it can extend to the skull base.

#19 · Acute Care and Diagnosis

How long can a cough from acute bronchitis reasonably last?

Answer

Up to three weeks. Acute bronchitis is a clinical diagnosis of cough with or without sputum in a patient without evidence of pneumonia.

#20 · Acute Care and Diagnosis

Should antibiotics be given for acute bronchitis with purulent sputum but normal vitals and a clear lung exam?

Answer

No. Purulent sputum reflects airway inflammation, not bacterial infection, and antibiotics do not shorten the illness. Symptomatic treatment is the correct answer.

#21 · Acute Care and Diagnosis

When is chest x-ray indicated in a patient with acute cough?

Answer

When pneumonia is suspected based on abnormal vital signs such as fever or tachypnea, or focal lung exam findings such as crackles, egophony, or dullness to percussion. Routine bronchitis does not need imaging.

#22 · Acute Care and Diagnosis

What does the CURB-65 score assess and what are its five criteria?

Answer

It estimates severity of community-acquired pneumonia and guides site of care. One point each for confusion, blood urea nitrogen greater than 19, respiratory rate 30 or higher, systolic blood pressure below 90 or diastolic below 60, and age 65 or older.

#23 · Acute Care and Diagnosis

A patient with CAP has a CURB-65 score of 0. Where should they be treated?

Answer

Outpatient. A score of 0 or 1 supports outpatient management with oral antibiotics and close follow-up.

#24 · Acute Care and Diagnosis

A patient with CAP has a CURB-65 score of 4. Where should they be treated?

Answer

Hospitalization is indicated, and a score of 4 or 5 raises the question of ICU level care given the high severity of illness.

#25 · Acute Care and Diagnosis

What is the first line outpatient antibiotic for a healthy patient with CAP and no comorbidities?

Answer

A macrolide, such as azithromycin, or doxycycline. This differs from patients with comorbidities who need broader coverage.

#26 · Acute Care and Diagnosis

What antibiotic regimen is preferred for an outpatient with CAP who has comorbidities like diabetes or COPD?

Answer

A respiratory fluoroquinolone, or a beta-lactam combined with a macrolide. Comorbidities increase risk of drug-resistant or atypical pathogens, so broader coverage is used.

#27 · Acute Care and Diagnosis

A patient with CAP is not improving after 72 hours of appropriate antibiotics. What should you consider?

Answer

Treatment failure, an alternate diagnosis, or a complication such as a parapneumonic effusion. Repeat evaluation, often with imaging, is warranted rather than simply waiting longer.

#28 · Acute Care and Diagnosis

What is the initial treatment for a moderate to severe asthma exacerbation?

Answer

A short-acting beta agonist combined with systemic corticosteroids. Peak flow or spirometry, when available, will show reduced values compared to the patient's personal best.

#29 · Acute Care and Diagnosis

What three cardinal features suggest antibiotics should be added to bronchodilators and steroids in a COPD exacerbation?

Answer

Increased dyspnea, increased sputum volume, and increased sputum purulence. Antibiotics are added when all three are present, or when only two are present as long as purulence is one of them. One feature alone, or two features without purulence, favors bronchodilators and steroids without antibiotics.

#30 · Acute Care and Diagnosis

Name four warning signs that mandate emergency department transfer during an asthma or COPD exacerbation.

Answer

Inability to speak in full sentences, use of accessory muscles, a silent chest on auscultation, and altered mental status or oxygen saturation that fails to improve with initial treatment.

#31 · Acute Care and Diagnosis

What exam findings help distinguish allergic rhinitis from an infectious cause?

Answer

Pale, boggy nasal turbinates and allergic shiners, which are dark circles under the eyes from venous congestion, along with a personal or family history of atopy, asthma, or eczema.

#32 · Acute Care and Diagnosis

What is the single most effective monotherapy for moderate to severe or persistent allergic rhinitis?

Answer

An intranasal corticosteroid. Second generation oral antihistamines are also first line but are generally less effective than intranasal steroids for persistent symptoms.

#33 · Acute Care and Diagnosis

What is the initial first aid step for anterior epistaxis?

Answer

Continuous direct pressure by pinching the soft cartilaginous part of the nose for ten to fifteen minutes while the patient leans forward, to avoid swallowing blood. Topical vasoconstrictors like oxymetazoline can be added if pressure alone fails.

#34 · Acute Care and Diagnosis

Where does most anterior epistaxis originate?

Answer

Kiesselbach's plexus, an anastomosis of blood vessels on the anterior nasal septum. This is the most common source of nosebleeds seen in ambulatory practice.

#35 · Acute Care and Diagnosis

What features suggest posterior rather than anterior epistaxis, and what is the management?

Answer

Heavier bleeding that is hard to control with anterior pressure, and bleeding from both nostrils or into the posterior pharynx. This usually requires ENT referral and posterior packing or arterial intervention.

#36 · Acute Care and Diagnosis

What underlying causes should be considered in a patient with recurrent or severe epistaxis?

Answer

Uncontrolled hypertension, anticoagulant or antiplatelet medication use, an underlying bleeding disorder, or, rarely, a nasal or sinus mass.

#37 · Acute Care and Diagnosis

What is the classic symptom triad of infectious mononucleosis?

Answer

Fever, pharyngitis, and posterior cervical or generalized lymphadenopathy. Profound fatigue and sometimes splenomegaly frequently accompany the triad.

#38 · Acute Care and Diagnosis

A patient treated with amoxicillin for presumed strep throat develops a diffuse rash. What diagnosis should you now suspect?

Answer

Infectious mononucleosis. A rash after amoxicillin or ampicillin in a patient with pharyngitis is a classic clue that the true diagnosis is EBV mononucleosis rather than strep.

#39 · Acute Care and Diagnosis

How reliable is the monospot test in the first week of infectious mononucleosis?

Answer

It can be falsely negative, especially in younger children. A peripheral smear showing atypical lymphocytes can support the diagnosis when suspicion remains high despite a negative monospot.

#40 · Acute Care and Diagnosis

How long should a patient with infectious mononucleosis avoid contact sports and heavy lifting, and why?

Answer

At least three to four weeks, due to the risk of splenic rupture. This precaution applies even if the spleen is not palpable on exam.

#41 · Acute Care and Diagnosis

What is the classic cough description in croup?

Answer

A barking, seal-like cough with inspiratory stridor and hoarseness, often worse at night. It typically affects children between six months and three years.

#42 · Acute Care and Diagnosis

What x-ray finding is associated with croup, and is imaging required?

Answer

The steeple sign from subglottic narrowing. Imaging is not required when the clinical picture is clear and is reserved for atypical presentations.

#43 · Acute Care and Diagnosis

How is mild croup without stridor at rest treated?

Answer

Cool mist or cool air exposure along with a single dose of oral dexamethasone to reduce airway inflammation.

#44 · Acute Care and Diagnosis

A child with croup has stridor at rest and retractions. What additional treatment is needed?

Answer

Nebulized racemic epinephrine in addition to steroids, followed by observation for two to four hours to watch for rebound symptoms as the epinephrine effect wears off.

#45 · Acute Care and Diagnosis

What are the three stages of pertussis in order?

Answer

The catarrhal stage, which looks like a mild cold and lasts one to two weeks and is when the patient is most contagious. The paroxysmal stage with severe coughing fits and an inspiratory whoop. The convalescent stage with gradual resolution over weeks to months.

#46 · Acute Care and Diagnosis

During which stage of pertussis is a patient most contagious but least likely to be correctly diagnosed?

Answer

The catarrhal stage, because it resembles a mild common cold with rhinorrhea and low grade fever, before the characteristic paroxysmal cough develops.

#47 · Acute Care and Diagnosis

What is the diagnostic test of choice for suspected pertussis, and when is it most sensitive?

Answer

PCR of a nasopharyngeal swab. It is most sensitive early in the illness, particularly during the catarrhal and early paroxysmal stages.

#48 · Acute Care and Diagnosis

What is the first line treatment for pertussis and who else should receive it?

Answer

A macrolide such as azithromycin. Close contacts, especially infants and pregnant women in their third trimester, should receive post-exposure prophylaxis with a macrolide regardless of their own vaccination status.

#49 · Acute Care and Diagnosis

Why is pertussis treatment still recommended even after the paroxysmal stage has started?

Answer

Because by that point antibiotics mainly limit spread to others rather than shorten the patient's own illness, since treatment is most effective at altering the disease course when given during the catarrhal stage.

#50 · Acute Care and Diagnosis

What is the appropriate management of a straightforward common cold in an adult?

Answer

Supportive care only: fluids, rest, saline nasal irrigation, and over-the-counter analgesics or decongestants as needed. Antibiotics are not indicated because the common cold is viral.

#51 · Acute Care and Diagnosis

How long can a cough persist after a common cold without indicating a new infection?

Answer

Up to three weeks. A lingering cough after resolution of other cold symptoms does not by itself mean a new or bacterial infection has developed.

#52 · Acute Care and Diagnosis

What does a watchful waiting approach for acute sinusitis involve?

Answer

Symptomatic treatment for seven days before starting antibiotics in patients with mild disease, since most cases of acute sinusitis resolve without antibiotic therapy.

#53 · Acute Care and Diagnosis

Why is otitis externa not treated with oral antibiotics as first line therapy?

Answer

Because it is a localized infection of the ear canal skin and soft tissue rather than a systemic or middle ear infection, so topical antibiotic-steroid drops are more effective and appropriate.

#54 · Acute Care and Diagnosis

What vital sign and exam findings raise suspicion for community-acquired pneumonia over simple bronchitis?

Answer

Fever, tachypnea, tachycardia, or focal lung findings such as crackles, egophony, or dullness to percussion. These findings should prompt a chest x-ray to confirm pneumonia.

#55 · Acute Care and Diagnosis

What triggers commonly precipitate an asthma exacerbation?

Answer

Upper respiratory infection, allergen exposure, or exercise. These common triggers help explain the acute onset of wheeze, chest tightness, and dyspnea.

#56 · Acute Care and Diagnosis

What smoking-related history is typical in a patient presenting with a COPD exacerbation?

Answer

A known diagnosis of COPD with a significant smoking history, presenting with increased dyspnea, increased sputum volume, and increased sputum purulence compared to baseline.

#57 · Acute Care and Diagnosis

Besides oral antihistamines and intranasal steroids, what other therapies help allergic rhinitis?

Answer

Intranasal antihistamines and leukotriene receptor antagonists as adjuncts or alternatives, and allergy testing with consideration of immunotherapy for symptoms that persist despite standard therapy.

#58 · Acute Care and Diagnosis

What CURB-65 score should prompt urgent escalation beyond routine outpatient pneumonia care?

Answer

A score of 2 or higher. This suggests need for a short inpatient stay or closely supervised care, and a score of 3 or higher points toward hospitalization.

#59 · Acute Care and Diagnosis

What is the overall theme for deciding between outpatient symptomatic care and escalation across acute respiratory and ENT complaints?

Answer

Most complaints are viral, self-limited, and respond to reassurance and symptomatic treatment, while specific validated criteria and red flags identify the smaller group of patients needing antibiotics, imaging, or urgent referral.

#60 · Acute Care and Diagnosis

What are the Ottawa Ankle Rules criteria for ordering an x-ray?

Answer

Pain in the malleolar zone plus one of: bone tenderness at the posterior edge or tip of the lateral malleolus, bone tenderness at the posterior edge or tip of the medial malleolus, or inability to bear weight for four steps both right after injury and in the clinic.

#61 · Acute Care and Diagnosis

What do the Ottawa Foot Rules add to the ankle rules?

Answer

Pain in the midfoot zone plus bone tenderness at the base of the fifth metatarsal, bone tenderness at the navicular, or inability to bear weight for four steps. They cover midfoot injuries separately from the ankle criteria.

#62 · Acute Care and Diagnosis

List the Ottawa Knee Rules criteria for imaging.

Answer

Age 55 or older, isolated tenderness of the patella, tenderness at the fibular head, inability to flex the knee to 90 degrees, or inability to bear weight for four steps. Any one criterion means an x-ray is indicated.

#63 · Acute Care and Diagnosis

How are ankle sprains graded?

Answer

Grade 1 is mild stretching with full weight bearing. Grade 2 is partial tearing with pain on weight bearing. Grade 3 is complete rupture with instability and inability to bear weight.

#64 · Acute Care and Diagnosis

Why is a Jones fracture treated differently from a fifth metatarsal avulsion fracture?

Answer

A Jones fracture occurs at the metaphyseal-diaphyseal junction, an area with poor blood supply, giving it a high nonunion risk. It usually needs orthopedic referral and non-weight-bearing care, unlike an avulsion fracture which heals well in a walking boot.

#65 · Acute Care and Diagnosis

Why do Salter-Harris fractures need extra caution in children?

Answer

They involve the growth plate and can disturb future bone growth. Any child with point tenderness over a growth plate after trauma needs imaging even if the film looks normal.

#66 · Acute Care and Diagnosis

List red flags for acute low back pain that warrant imaging.

Answer

Age over 50 with new pain, unexplained weight loss, cancer history, fever, IV drug use, immunosuppression, significant trauma, saddle anesthesia, new bowel or bladder dysfunction, and progressive neurologic deficit.

#67 · Acute Care and Diagnosis

What is cauda equina syndrome and how is it managed?

Answer

A surgical emergency with bilateral leg weakness, saddle anesthesia, and new urinary retention or incontinence. It requires immediate MRI and emergent neurosurgical or spine surgery consultation to prevent permanent deficit.

#68 · Acute Care and Diagnosis

What presentation suggests spinal epidural abscess rather than simple back pain?

Answer

Fever with severe back pain, sometimes with neurologic deficit, especially in patients with recent spinal procedures, IV drug use, or bacteremia. Fever plus new back pain should always raise this suspicion.

#69 · Acute Care and Diagnosis

What is the first line management for acute low back pain without red flags?

Answer

Activity as tolerated, short courses of NSAIDs, and reassurance that most episodes improve within four to six weeks. Bed rest is discouraged because it delays recovery.

#70 · Acute Care and Diagnosis

Describe the classic presentation of gout.

Answer

Rapid onset of intense pain, swelling, and erythema, most often at the first metatarsophalangeal joint (podagra). Attacks often start overnight and are triggered by alcohol, purine rich foods, dehydration, or diuretics.

#71 · Acute Care and Diagnosis

What crystals are seen in gout versus pseudogout on synovial fluid analysis?

Answer

Gout shows needle shaped, negatively birefringent monosodium urate crystals. Pseudogout shows rhomboid shaped, positively birefringent calcium pyrophosphate crystals.

#72 · Acute Care and Diagnosis

What findings should raise concern for septic arthritis?

Answer

An acutely hot, swollen, painful single joint with fever and severe restriction of range of motion with any attempted movement. Risk factors include prosthetic joints, IV drug use, immunosuppression, and recent joint injection.

#73 · Acute Care and Diagnosis

What synovial fluid studies should be sent when septic arthritis is suspected?

Answer

Cell count, gram stain, culture, and crystal analysis. Arthrocentesis is mandatory whenever septic arthritis is a concern.

#74 · Acute Care and Diagnosis

Can gout and septic arthritis occur in the same joint at the same time?

Answer

Yes. Finding crystals on aspiration does not rule out infection, particularly in patients with chronic tophaceous gout, so clinical suspicion should still guide further workup.

#75 · Acute Care and Diagnosis

What synovial white blood cell count strongly suggests joint infection?

Answer

Above roughly 50,000 per microliter strongly suggests infection, but lower counts do not exclude it. Septic arthritis should never be ruled out on cell count alone if suspicion remains high.

#76 · Acute Care and Diagnosis

What is reactive arthritis and what typically triggers it?

Answer

An asymmetric oligoarthritis of large weight bearing joints following a GI or GU infection by one to four weeks. Common triggers include Chlamydia, Salmonella, Shigella, Yersinia, and Campylobacter, sometimes with conjunctivitis and urethritis.

#77 · Acute Care and Diagnosis

What is the first line management for most tendinitis?

Answer

Activity modification, ice, and a structured stretching and strengthening program. Most tendinopathy responds to load management rather than prolonged rest.

#78 · Acute Care and Diagnosis

When should septic bursitis be suspected?

Answer

When a bursa is warm, red, and tender with fever. Aspiration is needed to distinguish infection from simple inflammatory bursitis before considering a steroid injection.

#79 · Acute Care and Diagnosis

How does cellulitis differ from an abscess on exam?

Answer

Cellulitis is spreading erythema, warmth, and tenderness without a distinct fluctuant collection. An abscess is a localized, fluctuant, pus filled pocket.

#80 · Acute Care and Diagnosis

What is the primary treatment for a skin abscess?

Answer

Incision and drainage. Antibiotics alone are not sufficient for a well formed abscess, though antibiotics are added when there is surrounding cellulitis, fever, immunosuppression, or a high risk location.

#81 · Acute Care and Diagnosis

What findings suggest necrotizing fasciitis rather than simple cellulitis?

Answer

Pain dramatically out of proportion to visible skin findings, rapid progression, skin necrosis, bullae, crepitus, or systemic toxicity. It requires emergent surgical debridement and broad spectrum antibiotics.

#82 · Acute Care and Diagnosis

What distribution pattern points to contact dermatitis from jewelry versus a plant?

Answer

Jewelry causes a rash confined to earlobes or a wrist. Plant contact, such as poison ivy, causes a linear, streaky rash where the plant brushed the skin.

#83 · Acute Care and Diagnosis

Describe classic psoriasis plaques and the Auspitz sign.

Answer

Well demarcated, erythematous plaques with silvery white scale on extensor surfaces, scalp, and lower back. The Auspitz sign is pinpoint bleeding when the scale is removed.

#84 · Acute Care and Diagnosis

What joint finding in a psoriasis patient suggests psoriatic arthritis?

Answer

New joint pain and swelling, sometimes with dactylitis, a diffusely swollen sausage digit. This combination should prompt evaluation for psoriatic arthritis.

#85 · Acute Care and Diagnosis

What is first line treatment for comedonal acne versus severe nodulocystic acne?

Answer

Comedonal acne is treated with topical retinoids first line. Severe nodulocystic acne that risks scarring may need oral isotretinoin, which requires pregnancy prevention counseling given its teratogenicity.

#86 · Acute Care and Diagnosis

Describe the classic appearance of tinea corporis.

Answer

An annular, scaling plaque with central clearing and an active, slightly raised border. A KOH prep showing branching hyphae supports the diagnosis.

#87 · Acute Care and Diagnosis

Why does tinea capitis require oral rather than topical antifungal therapy?

Answer

Topical agents do not penetrate the hair follicle adequately, so oral antifungal therapy is needed to clear the infection.

#88 · Acute Care and Diagnosis

What are the classic features of scabies and how are close contacts managed?

Answer

Intense pruritus worse at night, small papules, vesicles, and burrows in the finger webs, wrists, axillae, and groin. Household contacts should be treated at the same time even if asymptomatic, since incubation can take weeks.

#89 · Acute Care and Diagnosis

What is the first line treatment for scabies?

Answer

Topical permethrin applied from the neck down and left on for the recommended duration before washing off.

#90 · Acute Care and Diagnosis

Describe the classic presentation of herpes zoster.

Answer

A painful, vesicular rash in a dermatomal distribution that does not cross the midline, often preceded by pain or paresthesia for one to several days.

#91 · Acute Care and Diagnosis

When is antiviral therapy most effective for herpes zoster?

Answer

When started within 72 hours of rash onset. Early treatment reduces severity and duration of symptoms and lowers the risk of postherpetic neuralgia.

#92 · Acute Care and Diagnosis

What does Hutchinson's sign in herpes zoster indicate?

Answer

Zoster involving the tip of the nose suggests nasociliary nerve involvement and raises concern for ocular involvement, warranting urgent ophthalmology referral.

#93 · Acute Care and Diagnosis

What does the A in the ABCDE mnemonic for pigmented lesions stand for?

Answer

Asymmetry, meaning one half of the lesion does not match the other half. This is one feature that raises concern for melanoma.

#94 · Acute Care and Diagnosis

What does the B in ABCDE stand for and why does it matter?

Answer

Border, referring to irregular, notched, or poorly defined edges. An irregular border is a warning sign for melanoma rather than a benign nevus.

#95 · Acute Care and Diagnosis

What does the C and D in ABCDE stand for?

Answer

Color describes variation within a single lesion, including brown, black, red, white, or blue shades. Diameter greater than six millimeters raises concern, though smaller melanomas can occur.

#96 · Acute Care and Diagnosis

What does the E in ABCDE stand for?

Answer

Evolving, meaning any change in size, shape, color, or new symptoms such as bleeding or itching over time. Any lesion with several ABCDE features warrants biopsy or dermatology referral.

#97 · Acute Care and Diagnosis

Why should wounds be irrigated before closure?

Answer

Thorough irrigation with normal saline or tap water reduces bacterial contamination and lowers infection risk before the wound is closed.

#98 · Acute Care and Diagnosis

When should a wound be left open or closed in a delayed fashion instead of primarily closed?

Answer

Highly contaminated wounds, bite wounds, and wounds beyond the typical closure window (roughly six to eight hours on the extremities or trunk, longer on the face due to its excellent blood supply) should avoid primary closure, since it increases infection risk.

#99 · Acute Care and Diagnosis

When does a clean minor wound need a tetanus booster?

Answer

If more than ten years have passed since the last dose. For a dirty or high risk wound, that threshold drops to five years, and tetanus immune globulin may also be needed if vaccination history is incomplete.

#100 · Acute Care and Diagnosis

How does suture removal timing differ by body location?

Answer

Facial sutures are typically removed earlier, around five to seven days, to minimize scarring. Sutures over joints or high tension areas like the back stay in longer, around ten to fourteen days, to avoid dehiscence.

#101 · Acute Care and Diagnosis

Why are cat bites especially prone to infection?

Answer

Cat bites cause narrow, deep punctures that inoculate bacteria deeply while being difficult to irrigate adequately, raising infection risk compared to other bite types.

#102 · Acute Care and Diagnosis

What organism is associated with human bites and how are they typically managed?

Answer

Human bites carry risk of mixed flora including *Eikenella corrodens* and are generally treated with prophylactic antibiotics such as amoxicillin-clavulanate given the high infection risk.

#103 · Acute Care and Diagnosis

What is a closed fist injury or fight bite and why is it concerning?

Answer

It occurs when a clenched fist strikes teeth, causing a small wound that can hide joint capsule or tendon injury. It has a low threshold for surgical evaluation because damage is easy to underestimate on exam.

#104 · Acute Care and Diagnosis

What organism is classically associated with dog and cat bites?

Answer

Pasteurella multocida, which can cause rapid onset infection with significant redness and swelling within hours of the bite.

#105 · Acute Care and Diagnosis

What is the difference between a sprain and a strain?

Answer

A sprain is a ligament injury. A strain is an injury to muscle or tendon.

#106 · Acute Care and Diagnosis

What is PRICE therapy for acute soft tissue injuries?

Answer

Protection, rest, ice, compression, and elevation, followed by early functional mobilization once acute pain improves rather than prolonged immobilization.

#107 · Acute Care and Diagnosis

What common conditions cause lateral and medial elbow pain from overuse?

Answer

Lateral epicondylitis, known as tennis elbow, and medial epicondylitis, known as golfer's elbow. Both are diagnosed clinically by localized tenderness and pain with resisted movement.

#108 · Acute Care and Diagnosis

When is a corticosteroid injection appropriate for tendinitis or bursitis?

Answer

Once infection has been excluded through clinical evaluation or aspiration. A corticosteroid injection should never be given into an infected joint or bursa.

#109 · Acute Care and Diagnosis

Why is marking the border of cellulitis erythema with a pen useful?

Answer

It creates a visible reference point so the clinician can objectively assess whether the erythema is improving or worsening at a follow up visit.

#110 · Acute Care and Diagnosis

What antibiotic coverage differs between uncomplicated cellulitis and purulent cellulitis or abscess?

Answer

Uncomplicated nonpurulent cellulitis typically targets streptococci with an agent like cephalexin. Purulent cellulitis or abscess should also cover methicillin resistant *Staphylococcus aureus*.

#111 · Acute Care and Diagnosis

What vertebral condition should be considered after minor trauma in an older or osteoporotic patient with sudden severe back pain?

Answer

Vertebral compression fracture, which can occur even after a low energy fall from standing height in patients with osteoporosis risk factors.

#112 · Acute Care and Diagnosis

What is the key clinical clue that separates gout from an infected joint at the bedside?

Answer

There is no single clinical clue that reliably separates them, which is why aspiration with crystal analysis, gram stain, and culture is required whenever septic arthritis cannot be confidently excluded.

#113 · Acute Care and Diagnosis

What population is most classically affected by pseudogout?

Answer

Older adults, most commonly affecting the knee, from calcium pyrophosphate crystal deposition.

#114 · Acute Care and Diagnosis

What morphology terms describe primary skin lesions?

Answer

Macule, papule, plaque, vesicle, bulla, pustule, and nodule describe primary lesion types, while scale, crust, and ulceration describe secondary changes.

#115 · Acute Care and Diagnosis

Why is distribution important when evaluating a rash?

Answer

The pattern often points directly to a diagnosis, such as a symmetric rash on extensor surfaces suggesting psoriasis versus a dermatomal rash suggesting herpes zoster.

#116 · Acute Care and Diagnosis

What is the first clinical question to ask when evaluating any acute musculoskeletal or skin complaint?

Answer

Whether the patient has a red flag finding that changes management, before working through the more common differential diagnoses.

#117 · Acute Care and Diagnosis

What does right upper quadrant abdominal pain point toward?

Answer

The gallbladder and liver. Biliary colic and cholecystitis are top considerations, especially with fatty food intolerance, a positive Murphy sign, or known gallstones.

#118 · Acute Care and Diagnosis

What is the classic pain pattern of appendicitis?

Answer

Vague periumbilical pain that migrates to the right lower quadrant over 12 to 24 hours, with anorexia, low grade fever, and nausea. Point tenderness at McBurney point supports the diagnosis.

#119 · Acute Care and Diagnosis

Why can retrocecal appendicitis be easy to miss?

Answer

It can present with flank or back pain instead of the classic right lower quadrant pattern. Pelvic appendicitis can also mimic a UTI with urinary or rectal symptoms.

#120 · Acute Care and Diagnosis

What abdominal pain pattern raises concern for mesenteric ischemia?

Answer

Diffuse pain that is out of proportion to exam findings, especially in an older patient with atrial fibrillation or significant vascular disease. This needs urgent evaluation, not routine workup.

#121 · Acute Care and Diagnosis

What is the most common cause of acute viral gastroenteritis?

Answer

Norovirus and rotavirus are the most common viral causes. They are typically self-limited within a few days and managed with supportive care.

#122 · Acute Care and Diagnosis

Why are antibiotics usually avoided in bacterial gastroenteritis?

Answer

They can prolong carriage of the organism. With *E. coli* O157:H7 specifically, antibiotics can increase the risk of hemolytic uremic syndrome.

#123 · Acute Care and Diagnosis

A child develops bloody diarrhea after eating undercooked beef. What should you suspect?

Answer

E. coli O157:H7 infection with risk of hemolytic uremic syndrome. This is not routine gastroenteritis and should not be treated with antibiotics.

#124 · Acute Care and Diagnosis

What are alarm features that require upper endoscopy instead of empiric GERD treatment?

Answer

Dysphagia, odynophagia, unintended weight loss, gastrointestinal bleeding, iron deficiency anemia, or new symptoms in a patient over 60.

#125 · Acute Care and Diagnosis

How does duodenal ulcer pain differ from gastric ulcer pain?

Answer

Duodenal ulcer pain often improves with eating and returns a few hours later. Gastric ulcer pain often worsens with eating.

#126 · Acute Care and Diagnosis

What are the two most important causes of peptic ulcer disease?

Answer

Helicobacter pylori infection and chronic NSAID use. Both should be actively asked about in any patient with ulcer symptoms.

#127 · Acute Care and Diagnosis

What findings suggest a complicated peptic ulcer?

Answer

Melena or hematemesis suggests bleeding, sudden severe pain suggests perforation, and early satiety or vomiting suggests gastric outlet obstruction. Any of these needs urgent evaluation.

#128 · Acute Care and Diagnosis

How does appendicitis present differently in pregnancy?

Answer

The appendix is displaced upward and laterally as the uterus enlarges, so pain can present in the right upper quadrant or flank rather than the classic right lower quadrant location.

#129 · Acute Care and Diagnosis

Why is appendicitis often missed in older adults?

Answer

Their inflammatory response is often blunted, so they present with subtler, less localized pain and can progress to perforation before the diagnosis is suspected.

#130 · Acute Care and Diagnosis

What pain pattern suggests appendiceal perforation?

Answer

Pain that suddenly worsens then transiently improves, followed by diffuse peritonitis, fever, and a rigid abdomen.

#131 · Acute Care and Diagnosis

What distinguishes uncomplicated cystitis from pyelonephritis?

Answer

Cystitis presents with dysuria, frequency, urgency, and suprapubic discomfort without fever or flank pain. Pyelonephritis adds fever, chills, flank pain, and costovertebral angle tenderness.

#132 · Acute Care and Diagnosis

Which pyelonephritis patients need hospitalization for IV antibiotics?

Answer

Patients who are pregnant, appear toxic, cannot tolerate oral intake, or have signs of sepsis.

#133 · Acute Care and Diagnosis

What factors make a UTI complicated rather than uncomplicated?

Answer

Male sex, pregnancy, indwelling catheters, urinary tract abnormalities, immunosuppression, and recent instrumentation. These prompt a urine culture and longer treatment.

#134 · Acute Care and Diagnosis

When should asymptomatic bacteriuria be treated?

Answer

Only in pregnancy or before an invasive urologic procedure. Treating it in other patients does not improve outcomes and encourages antibiotic resistance.

#135 · Acute Care and Diagnosis

What is the classic presentation of renal colic?

Answer

Sudden, severe, colicky flank pain radiating toward the groin, often with the patient unable to find a comfortable position. Nausea, vomiting, and hematuria commonly accompany it.

#136 · Acute Care and Diagnosis

What is the standard imaging study for suspected nephrolithiasis?

Answer

Noncontrast CT is the most sensitive imaging study. Ultrasound is preferred as a first step in pregnant patients and children to avoid radiation.

#137 · Acute Care and Diagnosis

What red flags in nephrolithiasis require urgent urology involvement?

Answer

Fever suggesting an infected obstructing stone, a solitary kidney, bilateral obstruction, intractable pain or vomiting despite treatment, and acute kidney injury.

#138 · Acute Care and Diagnosis

Why is a fever with an obstructing kidney stone a urologic emergency?

Answer

It suggests an infected obstructed system. Prompt decompression along with antibiotics is needed, since antibiotics alone cannot clear the infection while the obstruction remains.

#139 · Acute Care and Diagnosis

How are chlamydia and gonorrhea treated?

Answer

Chlamydia is typically treated with doxycycline. Gonorrhea is treated with an injectable cephalosporin, with co-treatment for chlamydia considered given frequent coinfection.

#140 · Acute Care and Diagnosis

What is the diagnostic standard for chlamydia and gonorrhea?

Answer

Nucleic acid amplification testing on urine or genital swabs.

#141 · Acute Care and Diagnosis

What are the classic exam findings of pelvic inflammatory disease?

Answer

Lower abdominal pain, cervical motion tenderness, adnexal tenderness, and sometimes fever. It results from ascending infection, usually chlamydia or gonorrhea.

#142 · Acute Care and Diagnosis

What are the long term risks of untreated or repeated PID?

Answer

Infertility, chronic pelvic pain, and ectopic pregnancy.

#143 · Acute Care and Diagnosis

What is the primary syphilis lesion called?

Answer

A chancre, a painless genital ulcer.

#144 · Acute Care and Diagnosis

What is the classic presentation of secondary syphilis?

Answer

A diffuse rash that classically involves the palms and soles, along with systemic symptoms. It follows weeks after the primary chancre.

#145 · Acute Care and Diagnosis

What is the treatment of choice for syphilis at every stage?

Answer

Penicillin, with the dosing regimen depending on the stage of infection.

#146 · Acute Care and Diagnosis

What is the classic discharge and diagnostic finding of bacterial vaginosis?

Answer

A thin, grayish white discharge with a fishy odor. Diagnosis uses Amsel criteria or a wet mount showing clue cells, plus a positive whiff test.

#147 · Acute Care and Diagnosis

Is bacterial vaginosis classified as a sexually transmitted infection?

Answer

No. It is caused by an imbalance of normal vaginal flora rather than a single pathogen, even though it is associated with sexual activity.

#148 · Acute Care and Diagnosis

What is the classic presentation of candidal vaginitis?

Answer

Thick, white, cottage cheese like discharge with significant itching and vulvar irritation, often without a strong odor. Diagnosis is confirmed with pseudohyphae on wet mount.

#149 · Acute Care and Diagnosis

What is the classic presentation of trichomoniasis?

Answer

A frothy, yellow green, malodorous discharge with vulvovaginal irritation, sometimes with a strawberry appearing cervix. It is a true sexually transmitted infection caused by a protozoan.

#150 · Acute Care and Diagnosis

Why is trichomoniasis treated with oral rather than topical metronidazole?

Answer

Topical treatment does not reliably clear the infection from the urethra and periurethral glands, and sexual partners must be treated as well.

#151 · Acute Care and Diagnosis

How does migraine typically present?

Answer

Unilateral, throbbing pain with nausea and sensitivity to light and sound, sometimes preceded by a visual aura. Attacks often last hours to a day.

#152 · Acute Care and Diagnosis

How does tension headache differ from migraine?

Answer

Tension headache presents as bilateral, band like pressure without the nausea, light sensitivity, or throbbing quality typical of migraine, and is usually less disabling.

#153 · Acute Care and Diagnosis

What is the classic red flag headache presentation for subarachnoid hemorrhage?

Answer

Sudden, severe headache reaching maximum intensity within seconds to minutes, often called the worst headache of the patient's life. It requires urgent CT.

#154 · Acute Care and Diagnosis

What headache red flags suggest a secondary cause needing urgent evaluation?

Answer

Fever with neck stiffness, new neurologic deficits, worsening with Valsalva or in the morning, new headache after age 50, headache with cancer or immunosuppression, and headache after head trauma.

#155 · Acute Care and Diagnosis

What is the classic presentation of temporal arteritis?

Answer

New headache, jaw claudication, scalp tenderness, and visual symptoms in an older adult. Untreated, it can cause permanent vision loss, so high dose steroids are started urgently before biopsy confirmation.

#156 · Acute Care and Diagnosis

What is the first step in evaluating a patient with dizziness?

Answer

Determine whether the patient means true vertigo, a spinning sensation, versus presyncope, disequilibrium, or vague lightheadedness, since these point to different causes.

#157 · Acute Care and Diagnosis

What are the features of peripheral vertigo?

Answer

Intense vertigo often triggered or worsened by position change, commonly with nausea and vomiting but no other neurologic symptoms.

#158 · Acute Care and Diagnosis

How is benign paroxysmal positional vertigo diagnosed and treated?

Answer

Diagnosed with the Dix Hallpike maneuver and treated with canalith repositioning such as the Epley maneuver.

#159 · Acute Care and Diagnosis

How do vestibular neuritis and labyrinthitis differ?

Answer

Vestibular neuritis presents with continuous vertigo over days, often after a viral illness, without hearing loss. Labyrinthitis is similar but includes hearing loss.

#160 · Acute Care and Diagnosis

What is the classic triad of Meniere disease?

Answer

Episodic vertigo, hearing loss, and tinnitus, along with a sensation of ear fullness.

#161 · Acute Care and Diagnosis

What findings suggest central rather than peripheral vertigo?

Answer

Additional neurologic signs such as diplopia, dysarthria, or limb ataxia, vertical or direction changing nystagmus, and a HINTS exam favoring a central lesion. These need urgent stroke workup.

#162 · Acute Care and Diagnosis

What does the HINTS exam stand for and why does it matter?

Answer

Head Impulse, Nystagmus, and Test of Skew. It helps distinguish central vertigo, which can represent stroke, from benign peripheral vertigo.

#163 · Acute Care and Diagnosis

What is the key exam finding that distinguishes Bell's palsy from a central facial droop?

Answer

Bell's palsy involves the entire side of the face including the forehead. A droop that spares the forehead points to a central cause such as stroke.

#164 · Acute Care and Diagnosis

What other causes should be considered before diagnosing idiopathic Bell's palsy?

Answer

Lyme disease in endemic areas, Ramsay Hunt syndrome from herpes zoster with ear vesicles and pain, and rarely a parotid tumor if weakness is gradual rather than acute.

#165 · Acute Care and Diagnosis

What is the treatment for Bell's palsy and why does timing matter?

Answer

Oral corticosteroids are most effective when started within 72 hours of symptom onset. Eye protection is essential because impaired blinking risks corneal injury.

#166 · Acute Care and Diagnosis

What defines a transient ischemic attack?

Answer

Sudden onset of focal neurologic symptoms, such as unilateral weakness, numbness, slurred speech, or vision loss, that fully resolve, classically within an hour, without infarction on imaging.

#167 · Acute Care and Diagnosis

Why is a TIA considered a medical emergency?

Answer

The risk of stroke after a TIA is highest in the first 48 hours. It is a strong warning sign for an impending stroke, not a benign event to discharge without workup.

#168 · Acute Care and Diagnosis

What does the ABCD2 score estimate?

Answer

Short term stroke risk after a TIA, based on age, blood pressure, clinical features, duration of symptoms, and diabetes.

#169 · Acute Care and Diagnosis

What is used for secondary prevention after a TIA?

Answer

Antiplatelet therapy, high intensity statin therapy, and aggressive control of blood pressure, diabetes, and other vascular risk factors. Anticoagulation replaces antiplatelet therapy if atrial fibrillation is found.

#170 · Acute Care and Diagnosis

What features in seizure history support a true seizure over syncope?

Answer

A clear postictal confusion phase, along with a witnessed aura, movement pattern, loss of consciousness, tongue biting, and incontinence.

#171 · Acute Care and Diagnosis

What is the recommended neuroimaging for a first unprovoked seizure?

Answer

MRI, preferably, to look for a structural cause such as a tumor, prior stroke, or vascular malformation.

#172 · Acute Care and Diagnosis

How are provoked seizures managed differently from unprovoked seizures?

Answer

Provoked seizures, caused by a trigger such as fever, metabolic derangement, alcohol withdrawal, or medication effect, are managed by treating the underlying cause rather than starting long term antiseizure medication.

#173 · Acute Care and Diagnosis

What practical guidance is given after a first seizure besides medication decisions?

Answer

Driving restrictions and safety precautions such as avoiding swimming alone, since these carry real practical and legal implications for the patient.

#174 · Chronic Care Management

What blood pressure defines Stage 1 hypertension under current ACC/AHA guidelines?

Answer

Systolic 130 to 139 or diastolic 80 to 89. This is lower than the older JNC7 cutoffs, so do not confuse the two systems on the exam.

#175 · Chronic Care Management

What blood pressure defines Stage 2 hypertension?

Answer

Systolic 140 or higher, or diastolic 90 or higher. Medication is started immediately alongside lifestyle changes, usually with two drugs from different classes.

#176 · Chronic Care Management

When is drug therapy started in Stage 1 hypertension?

Answer

Only if the patient already has cardiovascular disease or a calculated ten year ASCVD risk of 10 percent or higher. Otherwise lifestyle modification alone is tried first.

#177 · Chronic Care Management

Why should ACE inhibitors and ARBs never be combined?

Answer

Combining them increases the risk of hyperkalemia and acute kidney injury without added cardiovascular benefit. Only one agent from this class should ever be used at a time.

#178 · Chronic Care Management

What is preferred first line antihypertensive therapy in Black patients without heart failure or CKD?

Answer

Thiazide diuretics or calcium channel blockers. These classes tend to work better than ACE inhibitors or ARBs alone as monotherapy in this population.

#179 · Chronic Care Management

What is the general blood pressure target for most adults, including many older adults?

Answer

Below 130/80. A less strict goal may be reasonable for frail patients or those with orthostatic symptoms.

#180 · Chronic Care Management

Which four groups automatically qualify for statin therapy regardless of a calculated risk score?

Answer

Patients with known ASCVD, patients with LDL 190 or higher, patients age 40 to 75 with diabetes, and patients age 40 to 75 without diabetes with a ten year ASCVD risk of 7.5 percent or higher.

#181 · Chronic Care Management

What LDL reduction does a high intensity statin achieve?

Answer

50 percent or more. High intensity statins such as atorvastatin 40 to 80 mg or rosuvastatin 20 to 40 mg are used for secondary prevention and for LDL 190 or higher.

#182 · Chronic Care Management

What is added if LDL remains above 70 on a maximally tolerated statin in a very high risk ASCVD patient?

Answer

Ezetimibe is added next. A PCSK9 inhibitor is considered if the LDL goal is still not met after ezetimibe.

#183 · Chronic Care Management

Is routine liver enzyme monitoring recommended after starting a statin?

Answer

No. Routine monitoring is no longer recommended unless there is a specific clinical concern, since clinically significant liver injury from statins is rare.

#184 · Chronic Care Management

How is HFrEF defined by ejection fraction?

Answer

Ejection fraction of 40 percent or less. This is the category with a proven four pillar mortality benefit regimen.

#185 · Chronic Care Management

How is HFpEF defined by ejection fraction?

Answer

Ejection fraction of 50 percent or higher, with 41 to 49 percent considered mildly reduced heart failure.

#186 · Chronic Care Management

What are the four pillars of guideline directed medical therapy for HFrEF?

Answer

An ARNI (or ACE inhibitor or ARB if ARNI is unavailable), a heart failure approved beta blocker, a mineralocorticoid receptor antagonist, and an SGLT2 inhibitor. All four are started and titrated together when tolerated.

#187 · Chronic Care Management

Do loop diuretics improve mortality in heart failure?

Answer

No. Loop diuretics help with symptom control and volume management but do not change mortality, unlike the four GDMT pillars.

#188 · Chronic Care Management

What is the key exam trap regarding HFpEF treatment?

Answer

There is no proven four drug mortality benefit pillar regimen for HFpEF like there is for HFrEF. Management centers on treating underlying causes, volume control, and SGLT2 inhibitors.

#189 · Chronic Care Management

Which drug class has proven benefit in HFpEF for reducing heart failure hospitalizations?

Answer

SGLT2 inhibitors, such as dapagliflozin or empagliflozin. This benefit applies in HFpEF as well as HFrEF.

#190 · Chronic Care Management

How should chronic heart failure patients be counseled about daily monitoring?

Answer

Track daily weights and report a gain of more than two to three pounds in a day immediately, since this often signals volume overload before other symptoms appear.

#191 · Chronic Care Management

What is the default rate control target in chronic atrial fibrillation?

Answer

A resting heart rate generally below 110, achieved with a beta blocker or non dihydropyridine calcium channel blocker.

#192 · Chronic Care Management

What score system determines the need for anticoagulation in atrial fibrillation?

Answer

The CHA2DS2-VASc score. It accounts for heart failure, hypertension, age, diabetes, prior stroke or TIA, vascular disease, and sex.

#193 · Chronic Care Management

At what CHA2DS2-VASc score is anticoagulation warranted?

Answer

A score of two or higher in men, or three or higher in women. A score of zero in men or one in women is low risk and anticoagulation is generally not needed.

#194 · Chronic Care Management

Why are DOACs generally preferred over warfarin in nonvalvular atrial fibrillation?

Answer

DOACs such as apixaban and rivaroxaban have a lower risk of intracranial bleeding and do not require routine INR monitoring compared to warfarin.

#195 · Chronic Care Management

In which atrial fibrillation patients is warfarin still required instead of a DOAC?

Answer

Patients with mechanical heart valves or moderate to severe mitral stenosis, since DOACs are not adequately studied or effective in these situations.

#196 · Chronic Care Management

What does the ABCDE mnemonic stand for in coronary artery disease secondary prevention?

Answer

Antiplatelet therapy, beta blocker, cholesterol management with a high intensity statin, control of blood pressure and diabetes, diet and daily activity, and education with smoking cessation.

#197 · Chronic Care Management

How long is dual antiplatelet therapy typically continued after acute coronary syndrome or stent placement?

Answer

Commonly six to twelve months, after which single antiplatelet therapy is usually continued long term.

#198 · Chronic Care Management

What statin intensity and LDL goal applies to all patients with known coronary artery disease?

Answer

A high intensity statin regardless of baseline LDL, with a goal of at least a 50 percent reduction from baseline.

#199 · Chronic Care Management

What is the general hemoglobin A1c goal for most nonpregnant adults with diabetes?

Answer

Below 7 percent. This is individualized, tighter for younger healthy patients and looser for older adults or those with limited life expectancy.

#200 · Chronic Care Management

What is first line pharmacologic therapy for type 2 diabetes?

Answer

Metformin plus lifestyle modification, unless contraindicated by significant renal impairment or acute illness.

#201 · Chronic Care Management

Down to what estimated GFR is metformin generally considered safe?

Answer

An estimated GFR of 30, with dose reduction considered below 45.

#202 · Chronic Care Management

What drives the choice of a second diabetes agent after metformin in current guidelines?

Answer

Comorbidities rather than A1c alone. Established ASCVD favors an SGLT2 inhibitor or GLP-1 agonist with proven cardiovascular benefit; heart failure or CKD favors an SGLT2 inhibitor; weight loss priority favors a GLP-1 agonist.

#203 · Chronic Care Management

What is the common exam trap regarding insulin initiation in type 2 diabetes?

Answer

Starting insulin too early. Guidelines favor optimizing a GLP-1 agonist or SGLT2 inhibitor first in a patient without severe symptomatic hyperglycemia.

#204 · Chronic Care Management

When should insulin be added in type 2 diabetes?

Answer

When there is severe hyperglycemia at diagnosis with symptoms like significant weight loss or ketosis, or when oral and injectable agents fail despite maximum tolerated doses.

#205 · Chronic Care Management

How often should a dilated eye exam be done for diabetic retinopathy screening?

Answer

Annually, though it can be extended to every two years in patients with well controlled diabetes and no retinopathy on a prior exam.

#206 · Chronic Care Management

What two tests screen for diabetic nephropathy and how often?

Answer

Annual urine albumin to creatinine ratio and estimated GFR. These detect early kidney involvement before symptoms appear.

#207 · Chronic Care Management

How is type 1 diabetes managed differently from type 2 from diagnosis?

Answer

Type 1 diabetes requires insulin from diagnosis, typically a basal bolus regimen or insulin pump, since these patients have absolute insulin deficiency.

#208 · Chronic Care Management

What lab pattern confirms primary hypothyroidism?

Answer

Elevated TSH with a low free T4, most commonly caused by Hashimoto thyroiditis in iodine sufficient regions.

#209 · Chronic Care Management

How is levothyroxine dosed and when is TSH rechecked after a dose change?

Answer

Roughly 1.6 micrograms per kilogram per day in most adults, started lower in older patients or those with coronary disease. TSH is rechecked six to eight weeks after any dose change.

#210 · Chronic Care Management

How should levothyroxine be taken relative to food and other medications?

Answer

On an empty stomach, ideally thirty minutes before breakfast, and separated from calcium, iron, and proton pump inhibitors, which impair its absorption.

#211 · Chronic Care Management

When is subclinical hypothyroidism treated?

Answer

When TSH is above 10, in pregnant patients, or in symptomatic patients with TSH between the upper normal limit and 10, based on clinical judgment.

#212 · Chronic Care Management

What lab pattern confirms hyperthyroidism, and what test confirms Graves disease specifically?

Answer

Suppressed TSH with elevated free T4 or T3 confirms hyperthyroidism. A positive TSH receptor antibody or diffusely increased radioactive iodine uptake confirms Graves disease.

#213 · Chronic Care Management

Why is methimazole generally preferred over propylthiouracil for Graves disease?

Answer

Methimazole has a better safety profile overall. Propylthiouracil is reserved for the first trimester of pregnancy and thyroid storm, where its lower teratogenic risk and faster onset matter.

#214 · Chronic Care Management

Why is radioactive iodine avoided in certain Graves disease patients?

Answer

It is avoided in pregnancy and in patients with significant Graves ophthalmopathy, since it can worsen eye disease.

#215 · Chronic Care Management

What two measurements does the KDIGO system use together to stage chronic kidney disease?

Answer

Estimated GFR category and albuminuria category. Either measurement alone underestimates risk, so both are needed for accurate staging.

#216 · Chronic Care Management

What GFR range defines CKD stage G3a and G3b?

Answer

G3a is GFR 45 to 59, and G3b is GFR 30 to 44. These midrange stages are where progression monitoring becomes especially important.

#217 · Chronic Care Management

What albuminuria level defines the A3 category?

Answer

Above 300 milligrams per gram. A1 is below 30, and A2 is 30 to 300.

#218 · Chronic Care Management

Can a patient have significant kidney disease with a normal GFR?

Answer

Yes. Elevated albuminuria alone can indicate meaningful kidney damage and risk even when GFR is still in the normal range.

#219 · Chronic Care Management

What two drug classes slow CKD progression, and in which patients?

Answer

ACE inhibitors or ARBs in patients with albuminuria, and SGLT2 inhibitors, which slow progression independent of diabetes status.

#220 · Chronic Care Management

At what serum bicarbonate level is oral sodium bicarbonate started in CKD, and why?

Answer

Below 22. Correcting metabolic acidosis in CKD can help slow further decline in kidney function.

#221 · Chronic Care Management

At what GFR is nephrology referral generally appropriate?

Answer

GFR below 30, or with rapidly declining GFR or difficult to control complications regardless of GFR stage.

#222 · Chronic Care Management

What BMI defines obesity, and what BMI defines overweight?

Answer

Obesity is BMI 30 or higher. Overweight is BMI 25 to 29.9. These cutoffs are lower in some populations such as Asian patients given higher metabolic risk at a given BMI.

#223 · Chronic Care Management

At what BMI is pharmacotherapy for obesity considered?

Answer

BMI 30 or higher, or BMI 27 or higher with a weight related comorbidity such as diabetes, hypertension, or obstructive sleep apnea.

#224 · Chronic Care Management

What class of medication has become first line pharmacotherapy for obesity?

Answer

GLP-1 receptor agonists and dual GLP-1/GIP agonists such as semaglutide and tirzepatide, given their strong weight loss efficacy and cardiometabolic benefits.

#225 · Chronic Care Management

At what BMI is bariatric surgery considered?

Answer

BMI 40 or higher, or 35 or higher with a significant weight related comorbidity, when other measures have not achieved adequate results.

#226 · Chronic Care Management

What is the most common side effect category with GLP-1 receptor agonists?

Answer

Gastrointestinal symptoms, such as nausea, vomiting, and constipation, which are the most frequent reason for dose adjustment or discontinuation.

#227 · Chronic Care Management

Why is obesity now managed as a chronic disease rather than a single intervention?

Answer

Because weight tends to be regained once treatment stops, so ongoing pharmacotherapy and lifestyle support are required for durable results.

#228 · Chronic Care Management

What compelling indication favors an ACE inhibitor or ARB regardless of blood pressure class?

Answer

Chronic kidney disease with albuminuria, or diabetes with albuminuria. These conditions favor ACE inhibitor or ARB use for kidney protection.

#229 · Chronic Care Management

How long are beta blockers continued after myocardial infarction?

Answer

At least a few years, and indefinitely if the patient also has heart failure with reduced ejection fraction.

#230 · Chronic Care Management

What secondary causes of hypertension should prompt further workup?

Answer

Young age, resistant hypertension, sudden onset, or suggestive exam findings, pointing toward causes like primary aldosteronism, renal artery stenosis, obstructive sleep apnea, CKD, or thyroid disease.

#231 · Chronic Care Management

Why is out of office blood pressure confirmation recommended before diagnosing hypertension?

Answer

White coat effect and masked hypertension are both common, and out of office readings via home or ambulatory monitoring give a more accurate picture that changes management.

#232 · Chronic Care Management

What population commonly develops anemia in CKD, and how is it treated?

Answer

CKD patients from reduced erythropoietin production. It is treated with iron repletion and sometimes an erythropoiesis stimulating agent once other causes of anemia are excluded.

#233 · Chronic Care Management

What drives secondary hyperparathyroidism in chronic kidney disease?

Answer

Impaired phosphorus excretion and reduced activation of vitamin D, which together are monitored with calcium, phosphorus, parathyroid hormone, and vitamin D levels.

#234 · Chronic Care Management

What screening exam detects diabetic peripheral neuropathy risk?

Answer

An annual comprehensive foot exam with monofilament testing, which identifies patients at risk for ulceration before it develops.

#235 · Chronic Care Management

What is the exam trap regarding CHA2DS2-VASc scoring?

Answer

Anchoring the anticoagulation decision on only age or only heart failure status rather than calculating the full score, since several smaller risk factors combined can cross the treatment threshold.

#236 · Chronic Care Management

What lifestyle measures are first line for every hypertensive patient regardless of stage?

Answer

The DASH diet, sodium reduction, weight loss, regular aerobic exercise, and limiting alcohol intake.

#237 · Chronic Care Management

What spirometry finding confirms a diagnosis of COPD?

Answer

A post-bronchodilator FEV1/FVC ratio below 0.70 in a patient with compatible symptoms and risk factors. Spirometry confirms the diagnosis and tracks decline over time, but it does not by itself decide which therapy to start.

#238 · Chronic Care Management

How does the GOLD ABE group classify a COPD patient with 2 moderate exacerbations last year, regardless of symptom score?

Answer

Group E. Two or more moderate exacerbations, or at least one hospitalization, in the prior year places a patient in Group E, which starts on LAMA-LABA combination therapy regardless of symptom burden.

#239 · Chronic Care Management

When is inhaled corticosteroid (ICS) added to COPD bronchodilator therapy?

Answer

ICS is added when exacerbations continue despite LAMA-LABA therapy, when blood eosinophils are 300 cells per microliter or higher, or when asthma-COPD overlap is suspected. ICS is not first-line monotherapy in COPD.

#240 · Chronic Care Management

What is the one COPD intervention proven to slow FEV1 decline and improve survival?

Answer

Smoking cessation. It is the single most effective intervention in the long-term management of COPD, more impactful than any inhaler regimen.

#241 · Chronic Care Management

When does long-term oxygen therapy improve survival in COPD?

Answer

Only in patients with resting hypoxemia, a PaO₂ of 55 mmHg or less, or 59 mmHg or less with cor pulmonale or polycythemia. Oxygen for mild hypoxemia alone does not improve survival.

#242 · Chronic Care Management

What is the preferred maintenance and reliever therapy for most steps of chronic asthma control today?

Answer

A combination ICS-formoterol inhaler used as both daily maintenance and as-needed reliever therapy, sometimes called SMART or MART therapy. It lowers exacerbation rates compared with a short-acting beta agonist used alone.

#243 · Chronic Care Management

Why is short-acting beta agonist monotherapy no longer recommended for asthma at any step?

Answer

SABA-only therapy without any inhaled corticosteroid is associated with higher exacerbation and mortality risk. Even mild persistent asthma should be paired with an ICS-containing inhaler.

#244 · Chronic Care Management

Before escalating asthma therapy for poor control, what two things should be checked first?

Answer

Inhaler technique and medication adherence. Poor technique is a common and reversible cause of apparent treatment failure, so it should be confirmed before adding medication.

#245 · Chronic Care Management

What screening tool is used to assess risk for obstructive sleep apnea, and what score suggests referral for a sleep study?

Answer

The STOP-BANG questionnaire. A score of 3 or higher suggests intermediate to high risk and warrants referral for polysomnography or home sleep apnea testing.

#246 · Chronic Care Management

How is OSA severity graded by AHI?

Answer

Mild is 5 to 14 events per hour, moderate is 15 to 29, and severe is 30 or more events per hour on the apnea-hypopnea index.

#247 · Chronic Care Management

What is first-line therapy for moderate to severe obstructive sleep apnea?

Answer

Continuous positive airway pressure (CPAP). It reduces daytime sleepiness, improves blood pressure control, and lowers cardiovascular risk when used consistently.

#248 · Chronic Care Management

What is a reasonable alternative to CPAP for mild OSA or CPAP intolerance?

Answer

A mandibular advancement device fitted by a dentist. It is generally less effective than CPAP at reducing AHI but is a reasonable option for select patients.

#249 · Chronic Care Management

How is chronic pain defined by duration?

Answer

Pain lasting longer than three months. Assessment should classify it as nociceptive, neuropathic, or mixed, and evaluate functional impact and psychosocial contributors.

#250 · Chronic Care Management

What are first-line medications for neuropathic pain?

Answer

Gabapentinoids such as gabapentin or pregabalin, and certain antidepressants such as duloxetine or tricyclic antidepressants like nortriptyline.

#251 · Chronic Care Management

Are opioids first-line therapy for chronic non-cancer pain?

Answer

No. Non-opioid and non-pharmacologic therapy comes first. Opioids are considered only after other options fail and require careful monitoring, including prescription drug monitoring program checks and periodic urine drug testing.

#252 · Chronic Care Management

Why should opioids and benzodiazepines not be co-prescribed?

Answer

Because of additive respiratory depression risk. Combining the two significantly increases the risk of fatal overdose.

#253 · Chronic Care Management

What is the correct approach when tapering a patient off long-term opioid therapy whose risks now outweigh benefits?

Answer

Taper slowly rather than stopping abruptly. Abrupt discontinuation increases risk of withdrawal and can push patients toward illicit opioid use.

#254 · Chronic Care Management

Is imaging required to diagnose osteoarthritis and start treatment?

Answer

No. Osteoarthritis is a clinical diagnosis in a patient over 45 with activity-related joint pain, morning stiffness under 30 minutes, and reduced range of motion. Radiographic severity often does not match symptom severity.

#255 · Chronic Care Management

Why are topical NSAIDs preferred over oral NSAIDs for knee and hand osteoarthritis when feasible?

Answer

Topical NSAIDs provide local relief with much less systemic absorption, lowering renal, cardiovascular, and gastrointestinal risk compared with oral NSAIDs.

#256 · Chronic Care Management

What is the concern with frequent intra-articular corticosteroid injections for osteoarthritis?

Answer

Repeated frequent injections have not shown long-term structural benefit and may accelerate cartilage loss with overuse, even though they can relieve short-term flares.

#257 · Chronic Care Management

Do glucosamine and chondroitin have proven benefit for osteoarthritis?

Answer

No. They lack consistent evidence of benefit and are not recommended as standard therapy for osteoarthritis.

#258 · Chronic Care Management

Which women should be screened for osteoporosis with DEXA?

Answer

All women 65 and older, and younger postmenopausal women whose risk factors raise fracture risk to that of a 65-year-old.

#259 · Chronic Care Management

What T-score defines osteoporosis versus osteopenia?

Answer

A T-score of negative 2.5 or lower defines osteoporosis. A T-score between negative 1.0 and negative 2.5 defines osteopenia, also called low bone mass.

#260 · Chronic Care Management

What tool estimates a patient's 10-year fracture risk using age, prior fracture, steroid use, and other factors?

Answer

FRAX. It calculates 10-year probability of major osteoporotic fracture and hip fracture, with or without a femoral neck T-score, and helps decide whether an osteopenic patient needs treatment.

#261 · Chronic Care Management

Can a patient with osteopenia, not osteoporosis, still qualify for pharmacologic treatment?

Answer

Yes. If FRAX-calculated 10-year risk reaches the treatment threshold, commonly 20% for major osteoporotic fracture or 3% for hip fracture, treatment is indicated even without an osteoporotic T-score.

#262 · Chronic Care Management

What are first-line agents for most patients starting osteoporosis treatment?

Answer

Oral bisphosphonates such as alendronate or risedronate, taken on an empty stomach with a full glass of water while remaining upright for 30 to 60 minutes to reduce esophageal irritation.

#263 · Chronic Care Management

What are two rare but important risks of long-term bisphosphonate therapy?

Answer

Osteonecrosis of the jaw and atypical femur fracture. These risks are why a drug holiday is considered after roughly 3 to 5 years in lower-risk patients who have responded well.

#264 · Chronic Care Management

What class of osteoporosis medication may be used first in patients with very high fracture risk?

Answer

Anabolic agents such as teriparatide or romosozumab, generally followed by an antiresorptive agent to maintain the bone density gain.

#265 · Chronic Care Management

What is the anchor first-line DMARD for rheumatoid arthritis?

Answer

Methotrexate, usually combined with folic acid supplementation to reduce gastrointestinal and hepatic side effects. It is contraindicated in pregnancy and significant liver disease.

#266 · Chronic Care Management

Why does early DMARD initiation matter in rheumatoid arthritis?

Answer

Early DMARD therapy prevents irreversible joint damage. Delay in starting DMARD therapy is associated with worse long-term outcomes.

#267 · Chronic Care Management

What should be screened before starting a biologic DMARD or JAK inhibitor for rheumatoid arthritis?

Answer

Latent tuberculosis and hepatitis B. These agents increase infection risk, so screening is required before initiation.

#268 · Chronic Care Management

How should chronic corticosteroids be used in rheumatoid arthritis management?

Answer

Sparingly, mainly as a bridge during flares while DMARD therapy takes effect. Long-term steroid use carries risks of osteoporosis, hyperglycemia, and cardiovascular disease.

#269 · Chronic Care Management

What is first-line pharmacotherapy for major depressive disorder and generalized anxiety disorder?

Answer

SSRIs, because of their favorable safety profile compared with older antidepressant classes. SNRIs are a reasonable first-line alternative, especially when pain symptoms coexist.

#270 · Chronic Care Management

How long should a patient be counseled to wait before an antidepressant trial is considered adequate?

Answer

4 to 6 weeks at an adequate dose. If there is no response by then, the next step is optimizing the dose further, switching agents, or augmenting therapy.

#271 · Chronic Care Management

Why is close follow-up needed in the first weeks of starting an antidepressant in a patient under 25?

Answer

Because of a boxed warning about increased suicidal thinking in patients under 25 during initial antidepressant treatment.

#272 · Chronic Care Management

Are benzodiazepines recommended for long-term maintenance therapy of generalized anxiety disorder?

Answer

No. They can relieve acute anxiety symptoms but are not recommended long term because of tolerance, dependence, and fall risk in older adults. An SSRI or SNRI plus psychotherapy is the correct maintenance choice.

#273 · Chronic Care Management

How long should antidepressant therapy continue after remission of a first depressive episode?

Answer

At least 6 to 12 months after remission to reduce relapse risk. Longer maintenance is considered after recurrent episodes.

#274 · Chronic Care Management

What validated tools track ongoing depression and anxiety control at follow-up visits?

Answer

The PHQ-9 for depression and the GAD-7 for anxiety. Both should be reassessed periodically along with suicidality screening in higher-risk patients.

#275 · Chronic Care Management

What three medications are approved for alcohol use disorder?

Answer

Naltrexone, acamprosate, and disulfiram. Naltrexone reduces craving and can start before abstinence, acamprosate is best started after abstinence, and disulfiram works through aversive deterrence.

#276 · Chronic Care Management

Why is naltrexone contraindicated in a patient with acute opioid use or on chronic opioid therapy?

Answer

It can precipitate opioid withdrawal or block needed pain control, since naltrexone is an opioid antagonist.

#277 · Chronic Care Management

What is considered first-line medication for opioid use disorder in most office-based primary care settings?

Answer

Buprenorphine, often combined with naloxone. It has a ceiling effect that lowers overdose risk and does not require enrollment in a certified opioid treatment program, unlike methadone.

#278 · Chronic Care Management

Why is methadone less accessible than buprenorphine for treating opioid use disorder in the United States?

Answer

Methadone is a full opioid agonist that must be dispensed through a certified opioid treatment program under strict regulation, unlike buprenorphine which can be prescribed in primary care.

#279 · Chronic Care Management

Why must a patient complete detoxification before starting long-acting injectable naltrexone for opioid use disorder?

Answer

Starting naltrexone too soon after opioid use precipitates acute withdrawal, since naltrexone blocks opioid receptors.

#280 · Chronic Care Management

Is counseling alone, without medication, the recommended first step for newly diagnosed opioid use disorder?

Answer

No. Medication for addiction treatment (MAT) is the evidence-based standard of care and reduces mortality more than counseling alone. It should be offered to essentially all patients with opioid use disorder.

#281 · Chronic Care Management

Who should be offered naloxone in the context of opioid use disorder management?

Answer

The patient with opioid use disorder and their household contacts, given naloxone's role in preventing fatal overdose.

#282 · Chronic Care Management

Does a relapse during MAT for opioid use disorder mean the medication should be discontinued?

Answer

No. Relapse is common in chronic substance use disorder and does not by itself represent treatment failure requiring discontinuation of medication.

#283 · Chronic Care Management

What age of symptom onset is required to diagnose adult ADHD?

Answer

Childhood-onset symptoms, typically before age 12, along with current impairment across more than one setting such as work and home.

#284 · Chronic Care Management

What is first-line pharmacotherapy for adult ADHD?

Answer

Stimulant medications, including methylphenidate and amphetamine-based agents. They produce the largest effect size for core ADHD symptoms.

#285 · Chronic Care Management

What should be assessed before starting a stimulant for adult ADHD?

Answer

Cardiovascular history, blood pressure, and heart rate, since stimulants raise heart rate and blood pressure modestly and carry a small increased cardiovascular risk.

#286 · Chronic Care Management

When are non-stimulant medications preferred for adult ADHD?

Answer

When stimulants are contraindicated, poorly tolerated, or there is significant concern about misuse potential, such as active or recent substance use disorder. Options include atomoxetine, extended-release guanfacine, and bupropion.

#287 · Chronic Care Management

What comorbid conditions commonly coexist with or mimic adult ADHD and should be screened for?

Answer

Mood disorders, anxiety, sleep disorders, and substance use disorder. These should be ruled out or treated alongside ADHD, since they can undermine treatment response.

#288 · Chronic Care Management

What percentage of predicted FEV1 defines GOLD 1 versus GOLD 4 COPD?

Answer

GOLD 1 (mild) is FEV1 80% or more of predicted.
GOLD 4 (very severe) is FEV1 under 30% of predicted.

#289 · Chronic Care Management

What defines well-controlled asthma at a routine follow-up visit?

Answer

Daytime symptoms twice a week or less, no nighttime awakenings, and rescue inhaler use twice a week or less.

#290 · Chronic Care Management

What common comorbidities can worsen asthma control if left untreated?

Answer

Allergic rhinitis and gastroesophageal reflux disease. Both should be addressed before escalating asthma medication for apparent poor control.

#291 · Chronic Care Management

What are two nonpharmacologic interventions offered to any COPD patient in GOLD group B or E?

Answer

Pulmonary rehabilitation and vaccination against influenza, pneumococcus, COVID-19, and, in older adults, respiratory syncytial virus.

#292 · Chronic Care Management

What lab monitoring is required for a patient on chronic methotrexate therapy?

Answer

Regular complete blood count and liver function tests, since methotrexate carries hematologic and hepatic toxicity risk.

#293 · Chronic Care Management

What antibodies support a diagnosis of rheumatoid arthritis alongside clinical findings?

Answer

Rheumatoid factor and anti-cyclic citrullinated peptide (anti-CCP) antibodies, along with elevated inflammatory markers and symmetric small joint swelling with morning stiffness over an hour.

#294 · Chronic Care Management

Why does rheumatoid arthritis management include cardiovascular risk screening?

Answer

Rheumatoid arthritis independently raises cardiovascular risk beyond traditional risk factors, so ongoing cardiovascular screening is part of chronic RA management.

#295 · Chronic Care Management

What vaccination precaution applies to rheumatoid arthritis patients on significant immunosuppression?

Answer

Live vaccines should be avoided in patients on significant immunosuppressive DMARD or biologic therapy.

#296 · Chronic Care Management

What is a key first step before starting long-term opioid therapy for chronic pain?

Answer

Check the state prescription drug monitoring program, establish a treatment agreement, and set functional goals rather than relying on pain-score-only goals.

#297 · Chronic Care Management

What is the recommended weight-related intervention for both osteoarthritis and obstructive sleep apnea?

Answer

Weight loss. It reduces joint load and symptoms in osteoarthritis and reduces OSA severity in patients with obesity, regardless of what device therapy is chosen for OSA.

#298 · Chronic Care Management

Why should total joint arthroplasty be reserved for later in osteoarthritis management?

Answer

It is reserved for patients with severe symptoms and functional limitation that persist despite optimized conservative management, since surgery carries risks that are not justified until conservative options are exhausted.

#299 · Chronic Care Management

What is the family physician's core longitudinal role across chronic pulmonary, musculoskeletal, and behavioral health conditions?

Answer

Confirming diagnosis with the correct objective test, matching therapy to validated severity or risk staging rather than symptoms alone, scheduling appropriate monitoring intervals, and coordinating specialist referral at defined trigger points.

#300 · Emergent and Urgent Care

How fast should an ECG be obtained for any adult presenting with chest pain?

Answer

Within 10 minutes of arrival. This determines whether the patient has a STEMI and drives all subsequent time-sensitive decisions.

#301 · Emergent and Urgent Care

What is the door to balloon time goal for STEMI at a PCI capable facility?

Answer

Under 90 minutes. If transfer to a PCI facility would take too long, fibrinolytics should be given with a door to needle goal under 30 minutes.

#302 · Emergent and Urgent Care

What ECG finding suggests a posterior MI and should be treated like a STEMI?

Answer

ST depression in leads V1 to V3 with a tall R wave. This is a STEMI equivalent even without classic ST elevation.

#303 · Emergent and Urgent Care

What is Wellens syndrome and why does it matter?

Answer

Deep symmetric T wave inversions or biphasic T waves in V2 to V3, signaling a critical proximal LAD lesion. Patients may be pain free when the ECG is taken, so it is easy to miss.

#304 · Emergent and Urgent Care

What are De Winter T waves?

Answer

Upsloping ST depression with tall symmetric T waves in the precordial leads, representing an acute LAD occlusion. It is another STEMI equivalent pattern.

#305 · Emergent and Urgent Care

Why should nitroglycerin be avoided in a patient with suspected right ventricular infarction?

Answer

RV infarction makes the patient preload dependent, so nitrates and diuretics can cause severe hypotension. Fluid boluses are the first line treatment instead.

#306 · Emergent and Urgent Care

How do you confirm right ventricular infarction on ECG?

Answer

Obtain a right sided ECG and look for ST elevation in lead V4R. It classically accompanies an inferior STEMI in leads II, III, and aVF.

#307 · Emergent and Urgent Care

What are the three components of the qSOFA score?

Answer

Respiratory rate 22 or higher, altered mentation, and systolic blood pressure 100 mmHg or lower. Two or more positive findings should raise concern for sepsis.

#308 · Emergent and Urgent Care

What must happen before antibiotics are given in suspected sepsis?

Answer

Blood cultures should be drawn, but this should not delay antibiotic administration. Broad spectrum antibiotics should be given within the first hour of recognition.

#309 · Emergent and Urgent Care

What is the fluid bolus recommendation in the sepsis hour-1 bundle?

Answer

30 mL per kg of crystalloid for hypotension or a lactate of 4 mmol per liter or higher, given rapidly within the first hour.

#310 · Emergent and Urgent Care

What is the first line vasopressor in septic shock and what MAP target is used?

Answer

Norepinephrine is first line, with a goal mean arterial pressure of 65 mmHg or above if the patient remains hypotensive after fluids.

#311 · Emergent and Urgent Care

What defines septic shock beyond sepsis alone?

Answer

Persistent hypotension requiring vasopressors plus a lactate above 2 mmol per liter despite adequate fluid resuscitation.

#312 · Emergent and Urgent Care

What is the first imaging study for a patient with sudden focal neurologic deficit?

Answer

A non-contrast head CT, to distinguish ischemic stroke from hemorrhagic stroke before any treatment decision is made.

#313 · Emergent and Urgent Care

What is the time window for IV alteplase in eligible ischemic stroke patients?

Answer

Up to 4.5 hours from clearly known symptom onset, with additional exclusion criteria applying in the 3 to 4.5 hour window such as age over 80 or use of oral anticoagulants.

#314 · Emergent and Urgent Care

What is the extended time window for mechanical thrombectomy in selected large vessel occlusion patients?

Answer

Up to 24 hours using perfusion imaging to identify salvageable brain tissue, which is longer than the thrombolysis window of 4.5 hours.

#315 · Emergent and Urgent Care

What blood pressure threshold must be met before giving alteplase for stroke?

Answer

Blood pressure must be controlled below roughly 185/110 mmHg. Severe uncontrolled hypertension above this level is an absolute contraindication to thrombolysis.

#316 · Emergent and Urgent Care

How is blood pressure managed differently in ischemic versus hemorrhagic stroke?

Answer

In ischemic stroke without thrombolytics, pressure is left permissive up to 220/120. In hemorrhagic stroke, pressure is lowered more aggressively, often to a systolic target around 140, to limit hematoma expansion.

#317 · Emergent and Urgent Care

Why should glucose always be checked in a patient with suspected stroke?

Answer

Hypoglycemia is a common and instantly reversible stroke mimic that can present with focal neurologic deficits.

#318 · Emergent and Urgent Care

What signs suggest intubation should not be delayed in acute respiratory failure?

Answer

Rapidly worsening mental status, inability to clear secretions, refractory hypoxemia despite maximal noninvasive support, and rising carbon dioxide with an abnormal respiratory rate.

#319 · Emergent and Urgent Care

When is noninvasive positive pressure ventilation a poor choice?

Answer

In a patient who cannot protect the airway, is vomiting, or is hemodynamically unstable. Intubation should not be delayed in these situations.

#320 · Emergent and Urgent Care

What is the diagnostic triad for diabetic ketoacidosis?

Answer

Hyperglycemia, ketosis, and metabolic acidosis with an elevated anion gap.

#321 · Emergent and Urgent Care

What is given first in DKA management, fluids or insulin?

Answer

Isotonic fluids are given first. Starting insulin before fluid resuscitation can worsen hypotension and cause potassium to shift into cells too quickly.

#322 · Emergent and Urgent Care

What potassium level requires holding insulin and replacing potassium first in DKA?

Answer

Below 3.3 mEq per liter. Giving insulin at this level can precipitate dangerous arrhythmias by lowering potassium further.

#323 · Emergent and Urgent Care

When should the insulin infusion be stopped in DKA treatment?

Answer

Only once the acidosis resolves, meaning the anion gap normalizes and bicarbonate improves, not simply when the glucose normalizes.

#324 · Emergent and Urgent Care

When should dextrose be added to IV fluids during DKA treatment?

Answer

Once glucose falls below roughly 200 mg per deciliter, to prevent hypoglycemia while the insulin infusion continues to correct the acidosis.

#325 · Emergent and Urgent Care

What criteria can be used to recognize anaphylaxis without a prior known allergy?

Answer

Acute onset with skin or mucosal involvement plus respiratory compromise or hypotension, or likely allergen exposure plus hypotension alone or two or more organ systems involved.

#326 · Emergent and Urgent Care

What is the first line treatment for anaphylaxis and where is it given?

Answer

Intramuscular epinephrine in the anterolateral thigh, given immediately rather than after antihistamines or steroids.

#327 · Emergent and Urgent Care

Why is delaying epinephrine to give diphenhydramine first a dangerous error in anaphylaxis?

Answer

Antihistamines do not treat the airway swelling or hypotension that actually kills patients in anaphylaxis, so they should never replace or precede epinephrine.

#328 · Emergent and Urgent Care

What alternative treatment may be needed for anaphylaxis in a patient on beta blockers?

Answer

Glucagon, because these patients may not respond well to epinephrine and glucagon works through a different receptor pathway.

#329 · Emergent and Urgent Care

Why should patients be observed for several hours after treated anaphylaxis?

Answer

Because a biphasic reaction, a second wave of symptoms without re-exposure to the trigger, can occur hours later.

#330 · Emergent and Urgent Care

What is the initial priority in a significant GI bleed, resuscitation or diagnosis?

Answer

Resuscitation first. Two large bore IVs, type and crossmatch, and fluid or blood product resuscitation come before endoscopy in an unstable patient.

#331 · Emergent and Urgent Care

What two treatments should be started before endoscopy in suspected variceal bleeding?

Answer

Octreotide and prophylactic antibiotics, since both reduce mortality and rebleeding risk in cirrhotic patients.

#332 · Emergent and Urgent Care

Why is aggressive crystalloid resuscitation risky in variceal GI bleeding?

Answer

Over-resuscitation can raise portal pressure and worsen bleeding, so a more restrictive transfusion strategy is generally preferred once major instability is addressed.

#333 · Emergent and Urgent Care

What hemoglobin threshold is generally used for transfusion in stable GI bleed patients?

Answer

Around 7 grams per deciliter, though a higher threshold may be used in patients with active cardiac disease or ongoing brisk bleeding.

#334 · Emergent and Urgent Care

What does the mnemonic AEIOU TIPS cover in the workup of altered mental status?

Answer

Alcohol, Epilepsy or Electrolytes, Infection, Overdose or Oxygen, Uremia, Trauma or Temperature, Insulin or metabolic causes, Psychiatric causes, and Stroke or Shock.

#335 · Emergent and Urgent Care

What are the first three actions for any patient with altered mental status?

Answer

A rapid glucose check, pulse oximetry, and a focused neurologic exam, since hypoglycemia and hypoxia are instantly reversible causes.

#336 · Emergent and Urgent Care

Why is infection an important cause of altered mental status in older adults specifically?

Answer

Infection, especially UTI or pneumonia, frequently causes confusion in the elderly even without fever, sometimes called silent sepsis in this population.

#337 · Emergent and Urgent Care

How is delirium distinguished from dementia?

Answer

By its acute onset and fluctuating course. Assuming chronic dementia in a patient with a genuinely new acute change can cause a serious treatable cause to be missed.

#338 · Emergent and Urgent Care

What red flags in a syncope history suggest a cardiac cause?

Answer

Syncope during exertion, syncope while supine, chest pain or palpitations before the event, family history of sudden cardiac death, an abnormal ECG, or known structural heart disease.

#339 · Emergent and Urgent Care

What ECG findings raise concern for an arrhythmic cause of syncope?

Answer

Prolonged QT interval, a Brugada pattern, or evidence of a prior myocardial infarction.

#340 · Emergent and Urgent Care

What three things should every patient with syncope receive during initial evaluation?

Answer

An ECG, orthostatic vital signs, and a careful history of events immediately before and after the episode.

#341 · Emergent and Urgent Care

Why is syncope without any warning symptoms in an older patient more concerning?

Answer

Absence of a prodrome in an older patient is more suggestive of an arrhythmic cause rather than a benign vasovagal episode.

#342 · Emergent and Urgent Care

What does the ABCDE primary survey stand for in trauma assessment?

Answer

Airway with cervical spine protection, Breathing, Circulation with hemorrhage control, Disability using a rapid neurologic check, and Exposure with temperature control.

#343 · Emergent and Urgent Care

How is a tension pneumothorax treated in the primary survey?

Answer

With immediate needle or finger decompression, without waiting for imaging to confirm the diagnosis first.

#344 · Emergent and Urgent Care

How should an open pneumothorax be managed?

Answer

With a three sided occlusive dressing, which allows air to escape but prevents air from being drawn back into the chest.

#345 · Emergent and Urgent Care

Why can young trauma patients appear falsely stable before sudden decompensation?

Answer

They can physiologically compensate for significant blood loss until they suddenly cannot, so shock must be recognized early using vital signs, skin findings, and mental status rather than waiting for hypotension.

#346 · Emergent and Urgent Care

When should the primary survey be repeated in trauma care?

Answer

Whenever the patient deteriorates. The ABCDE sequence should be run again from the beginning rather than jumping straight to a single suspected cause.

#347 · Emergent and Urgent Care

What comes after the primary survey is complete in trauma care?

Answer

The secondary survey, a more detailed head to toe examination and history, performed only after all immediately life threatening problems have been identified and treated.

#348 · Emergent and Urgent Care

What sequencing trap is common across sepsis, DKA, and anaphylaxis management?

Answer

Blood cultures before antibiotics in sepsis, fluids before insulin in DKA, and epinephrine before antihistamines in anaphylaxis. Order of treatment matters as much as the treatment itself.

#349 · Emergent and Urgent Care

What is the general management principle for right ventricular infarction that differs from typical inferior MI care?

Answer

These patients are preload dependent, so nitrates and diuretics are avoided and fluid boluses are used instead to maintain adequate right ventricular filling.

#350 · Emergent and Urgent Care

What chest pain features in an outpatient clinic mandate calling EMS instead of having the patient drive to the ED?

Answer

Syncope with chest pain, signs of new heart failure, hypotension, and pain worsening despite rest. These suggest an unstable cardiac event needing immediate transport and monitoring, not self-transport.

#351 · Emergent and Urgent Care

What chest pain history should make you think aortic dissection instead of ACS?

Answer

Tearing or ripping pain radiating to the back, a pulse or blood pressure difference between the arms, and a widened mediastinum on chest x-ray. Giving aspirin or anticoagulants here is dangerous and wrong.

#352 · Emergent and Urgent Care

What is the fever cutoff and required workup for an infant younger than 28 days old?

Answer

A rectal temperature of 100.4 Fahrenheit or higher requires a full sepsis evaluation: blood culture, catheterized urine culture, and cerebrospinal fluid analysis, plus empiric antibiotics and admission. Sending home is never correct.

#353 · Emergent and Urgent Care

How is a febrile infant 29 to 60 days old generally managed?

Answer

Still considered high risk. Even with newer low-risk criteria available, the safe default is a full or near-full evaluation with strong consideration of admission, especially if inflammatory markers are elevated or the infant looks ill.

#354 · Emergent and Urgent Care

A fully vaccinated, well-appearing toddler has fever without an identifiable source. What is the appropriate workup?

Answer

Generally no aggressive workup is needed. Pneumococcal and Hib vaccines have dramatically reduced occult bacteremia risk in this group, unlike unvaccinated or under-vaccinated children who need a lower threshold for labs.

#355 · Emergent and Urgent Care

Name pediatric fever red flags that push toward emergent transfer regardless of age.

Answer

Lethargy, poor feeding, decreased urine output, petechial rash, and a bulging fontanelle in an infant. These indicate the child is ill-appearing, which matters more than the fever number itself.

#356 · Emergent and Urgent Care

What peak flow percentage and clinical signs indicate a severe asthma exacerbation needing emergency transfer?

Answer

Peak flow below 40 percent of predicted, inability to speak in full sentences, accessory muscle use, and a silent chest on auscultation. Oxygen saturation below 90 percent or altered mental status are also ominous.

#357 · Emergent and Urgent Care

Why can a suddenly calmer-looking asthma patient be a warning sign rather than reassurance?

Answer

A patient who was working hard to breathe and then appears to relax without objective improvement may actually be tiring out and heading toward respiratory failure, not truly improving.

#358 · Emergent and Urgent Care

Why can pulse oximetry be misleading in a COPD exacerbation?

Answer

COPD patients who chronically retain carbon dioxide can have falsely reassuring oxygen saturation. Mental status and work of breathing must be assessed directly rather than relying on pulse ox alone.

#359 · Emergent and Urgent Care

Within what time window can a clean, minor wound generally be closed primarily?

Answer

Within 6 to 8 hours of injury, though this window can extend on the face due to its excellent blood supply. Heavily contaminated wounds, bites, or wounds over 12 to 24 hours old are usually left open.

#360 · Emergent and Urgent Care

A patient with a clean minor wound and 3 or more prior tetanus doses needs a booster if how much time has passed?

Answer

More than 10 years since the last dose. For a dirty or tetanus-prone wound, that threshold tightens to more than 5 years.

#361 • Emergent and Urgent Care

A patient with an unknown or incomplete tetanus vaccination history (fewer than 3 doses) has a dirty, tetanus-prone wound. What is needed?

Answer

Both tetanus toxoid vaccine and tetanus immune globulin, given at separate injection sites.

#362 • Emergent and Urgent Care

Which animal bite type carries a particularly high infection risk and often needs prophylactic antibiotics?

Answer

Cat bites and hand wounds, typically treated with amoxicillin-clavulanate. Human bites also carry high polymicrobial infection risk and are generally not closed primarily.

#363 · Emergent and Urgent Care

What diagnostic criteria define anaphylaxis rather than a localized allergic reaction?

Answer

Acute onset with skin or mucosal involvement plus respiratory compromise or low blood pressure, OR two or more organ systems involved after allergen exposure, OR hypotension after exposure to a known allergen for that patient.

#364 · Emergent and Urgent Care

What is the first-line treatment for anaphylaxis and why should it never be delayed?

Answer

Intramuscular epinephrine in the anterolateral thigh, given immediately. It is the only agent proven to reverse the airway and cardiovascular collapse of anaphylaxis; antihistamines and steroids are adjuncts, not substitutes.

#365 · Emergent and Urgent Care

Why must every anaphylaxis patient be transported for observation even after apparent full recovery?

Answer

Biphasic reactions can occur hours later, typically 4 to 12 hours after the initial episode, sometimes even later. Sending the patient home directly from clinic is unsafe.

#366 · Emergent and Urgent Care

What are the clinical signs of moderate dehydration (roughly 6 to 9 percent body weight loss) in a child?

Answer

Sunken eyes, decreased skin turgor, delayed capillary refill, and reduced urine output. A trial of oral rehydration in clinic with reassessment is appropriate, sometimes aided by ondansetron if vomiting limits intake.

#367 · Emergent and Urgent Care

What signs indicate severe pediatric dehydration requiring emergency IV or intraosseous fluids?

Answer

Lethargy, sunken fontanelle in infants, capillary refill greater than 3 seconds, cool mottled extremities, and minimal or absent urine output. Oral rehydration alone is unsafe at this severity.

#368 · Emergent and Urgent Care

Why is hypotension considered a late and worrisome finding in a dehydrated or shocked child?

Answer

Children compensate well physiologically, so tachycardia out of proportion to fever often precedes blood pressure drops. By the time hypotension appears, the child may already be in decompensated shock.

#369 · Emergent and Urgent Care

What exam findings distinguish acute angle-closure glaucoma from simple conjunctivitis?

Answer

Angle-closure glaucoma causes severe eye pain, headache, nausea and vomiting, a fixed mid-dilated pupil, a hazy cornea, and markedly elevated intraocular pressure. Conjunctivitis has redness and discharge but normal pupil, normal vision, and no severe pain.

#370 · Emergent and Urgent Care

What is the classic presentation and key exam finding of central retinal artery occlusion?

Answer

Sudden, painless, profound monocular vision loss described as a curtain coming down, with a pale retina and cherry-red spot at the macula on exam. Treatment window is narrow, generally under 4 to 6 hours.

#371 · Emergent and Urgent Care

How do you distinguish orbital cellulitis from preseptal cellulitis?

Answer

Orbital cellulitis has pain with eye movement, proptosis, decreased visual acuity, and restricted extraocular movements, requiring IV antibiotics and CT imaging. Preseptal cellulitis is confined to the eyelid without these orbital signs and can often be treated orally as an outpatient.

#372 · Emergent and Urgent Care

What does Hutchinson's sign indicate in a patient with facial vesicles?

Answer

Vesicles on the tip of the nose in herpes zoster ophthalmicus suggest nasociliary nerve involvement, indicating a high risk of ocular complications and the need for urgent ophthalmology evaluation.

#373 · Emergent and Urgent Care

What is the immediate first step for any chemical exposure to the eye, before history or exam?

Answer

Immediate copious irrigation. This takes priority over obtaining a detailed history or performing a full exam, since delay worsens tissue damage.

#374 · Emergent and Urgent Care

What physical exam findings suggest testicular torsion rather than epididymitis?

Answer

A high-riding, horizontally lying testicle and an absent cremasteric reflex on the affected side, with diffuse tenderness. Epididymitis usually has a normal-lying testicle and a positive cremasteric reflex.

#375 · Emergent and Urgent Care

Why is the Prehn sign unreliable in evaluating suspected testicular torsion?

Answer

Relief of pain with testicular elevation is classically linked to epididymitis, but it is inconsistent and should never be used to rule out torsion, since testicular salvage depends on rapid, accurate diagnosis.

#376 · Emergent and Urgent Care

What are the approximate testicular salvage rates at 6, 12, and beyond 24 hours after torsion onset?

Answer

Greater than 90 percent within 6 hours, dropping to near 50 percent by 12 hours, and very low beyond 24 hours. This narrow window is why suspected torsion needs immediate urology consultation.

#377 · Emergent and Urgent Care

Why should clinicians avoid waiting for ultrasound confirmation when torsion is strongly suspected clinically?

Answer

A normal or equivocal ultrasound does not exclude torsion, especially intermittent torsion, and should never override strong clinical suspicion. Delaying surgery for imaging risks losing the testicle.

#378 · Emergent and Urgent Care

Why is a button battery in the ear or nose considered the most urgent foreign body scenario?

Answer

It can cause liquefactive tissue necrosis within hours through electrical current and chemical burn. Emergent removal, generally within 2 hours if possible, is required, and observation is never appropriate.

#379 · Emergent and Urgent Care

How should a live insect in the ear canal be managed before removal?

Answer

Kill it first, typically by instilling mineral oil or lidocaine into the canal, before attempting extraction. This reduces pain and struggling during removal.

#380 · Emergent and Urgent Care

What nasal foreign bodies or patient factors should prompt ENT referral instead of clinic removal attempts?

Answer

Objects that are round, smooth, or posteriorly located, or an uncooperative child at risk of the object being pushed toward the airway. Repeated forceful attempts in clinic risk aspiration.

#381 · Emergent and Urgent Care

What eye exam findings suggest a penetrating globe injury requiring emergency ophthalmology referral?

Answer

An irregularly shaped pupil, visible prolapsed uveal tissue, or a positive Seidel sign showing aqueous leakage. Use an eye shield, not a pressure patch, and avoid further manipulation.

#382 · Emergent and Urgent Care

What is the first step when evaluating any painful red eye for a corneal foreign body?

Answer

Fluorescein staining to assess for a corneal abrasion, plus eyelid eversion to check for a hidden foreign body under the upper lid.

#383 · Emergent and Urgent Care

What three questions form the triage framework for urgent ambulatory presentations?

Answer

Is there an unstable airway, breathing, or circulation problem. Is there a narrow time window in which intervention changes outcome. Does the patient fall into a high-risk category by age or comorbidity.

#384 · Emergent and Urgent Care

Give three examples of conditions with a narrow, well-defined treatment time window discussed in this chapter.

Answer

Testicular torsion (testicle salvage drops sharply after 6 hours), central retinal artery occlusion (under 4 to 6 hours), and button battery ingestion or insertion (ideally removed within 2 hours).

#385 · Emergent and Urgent Care

What ECG or exam finding on a suspected PE workup should prompt use of a formal risk score rather than reassurance alone?

Answer

Pleuritic chest pain after a recent long flight or surgery. Wells criteria plus D-dimer or imaging should guide the decision, not clinical reassurance alone.

#386 · Emergent and Urgent Care

What is the recommended dose and route of aspirin for suspected ACS in clinic while awaiting EMS?

Answer

325 mg chewed, given while awaiting transport, as long as there is no contraindication such as an active bleeding disorder or true aspirin allergy.

#387 · Emergent and Urgent Care

In a peak flow assessment for asthma, what percentage of predicted value indicates the patient has improved enough for possible outpatient management?

Answer

Above 70 percent of predicted, combined with a period of observation without recurrence of symptoms after standard bronchodilator therapy.

#388 · Emergent and Urgent Care

Why are human bite wounds generally not closed primarily?

Answer

They carry a significant risk of polymicrobial infection. Delayed closure or leaving the wound open reduces the risk of trapping bacteria under a sutured wound.

#389 · Emergent and Urgent Care

What adjunct medication can improve success of oral rehydration therapy in a vomiting child?

Answer

Ondansetron. It can reduce vomiting enough to allow successful oral rehydration and avoid the need for IV fluids in moderately dehydrated children.

#390 · Emergent and Urgent Care

What percentage of body weight loss defines mild pediatric dehydration, and how is it managed?

Answer

Roughly 3 to 5 percent, presenting with slightly dry mucous membranes and normal vital signs. It can be managed with oral rehydration solution at home and close follow-up instructions.

#391 · Emergent and Urgent Care

Why should a patient with suspected acute angle-closure glaucoma not simply be treated as conjunctivitis?

Answer

Angle-closure glaucoma causes permanent vision loss within hours if untreated. The severe pain, fixed mid-dilated pupil, and elevated intraocular pressure distinguish it and require immediate ophthalmologic referral.

#392 · Emergent and Urgent Care

Why does central retinal artery occlusion warrant urgent systemic workup even beyond the treatment window for vision?

Answer

It represents an ocular stroke and a marker of systemic vascular disease, so urgent cardiovascular and stroke workup is needed regardless of whether vision can still be salvaged.

#393 · Emergent and Urgent Care

What imaging is used to evaluate for orbital cellulitis and possible abscess?

Answer

CT imaging of the orbit, along with IV antibiotics and often ophthalmology or ENT consultation, because untreated infection can spread to the cavernous sinus or intracranial space.

#394 · Emergent and Urgent Care

What historical and exam features suggest epididymitis rather than testicular torsion?

Answer

A more gradual onset, a normal-lying testicle, a positive cremasteric reflex, and often urinary symptoms or known sexually transmitted infection risk. It can be managed with outpatient antibiotics.

#395 · Emergent and Urgent Care

What techniques can be tried before instrumentation for a cooperative child with a nasal foreign body?

Answer

Having the child blow their nose forcefully while occluding the unaffected nostril, or using positive pressure from a parent's breath into the child's mouth, sometimes called the parent's kiss technique.

#396 · Emergent and Urgent Care

What does a rust ring around a corneal foreign body indicate and what follow-up may be needed?

Answer

It indicates a retained metallic foreign body that has begun to oxidize. Referral for ring removal may be needed if it cannot be fully cleared in clinic.

#397 · Emergent and Urgent Care

Why should reproducible chest wall tenderness on palpation not fully reassure a clinician about cardiac risk?

Answer

It lowers but does not eliminate the probability of cardiac disease, especially in older patients or those with cardiac risk factors, so it should not be used alone to rule out ACS.

#398 · Emergent and Urgent Care

What should be prescribed and taught before discharging a patient after an anaphylaxis episode?

Answer

An epinephrine auto-injector with proper technique training, along with a referral to allergy for further evaluation and management.

#399 · Emergent and Urgent Care

What is the general rule for deciding outpatient management versus emergency transfer across all urgent presentations in this chapter?

Answer

If there is airway, breathing, or circulation instability, a narrow time-critical window, or a high-risk age or comorbidity category, emergency transfer is the safe answer. If all three are clearly absent and follow-up is reliable, outpatient management with return precautions is appropriate.

#400 · Preventive Care

What does a USPSTF Grade A recommendation mean?

Answer

There is high certainty of substantial net benefit. The service should be offered to eligible patients. Grade A and B services must be covered by most insurance plans with no cost sharing under the ACA.

#401 · Preventive Care

What does a USPSTF Grade C recommendation mean?

Answer

The net benefit is small, so the service should be offered selectively based on shared decision making with the patient. It is not a blanket recommendation for or against the service.

#402 · Preventive Care

What does a USPSTF Grade D recommendation mean?

Answer

The task force recommends against offering the service because there is moderate or high certainty of no net benefit, or that harms outweigh benefits. PSA screening in men 70 and older is a Grade D example.

#403 · Preventive Care

What does a USPSTF Grade I statement mean?

Answer

Current evidence is insufficient to assess the balance of benefits and harms. No recommendation can be made either for or against the service based on the available evidence.

#404 · Preventive Care

At what age does the USPSTF recommend starting biennial mammography?

Answer

Age 40, continuing through age 74. This is a Grade B recommendation that was lowered from age 50 in 2024. Evidence is insufficient for women 75 and older.

#405 · Preventive Care

How often should average risk women get mammograms between ages 40 and 74?

Answer

Every two years, known as biennial screening. This is the USPSTF Grade B recommendation for the general population without elevated breast cancer risk factors.

#406 · Preventive Care

Who should be referred for BRCA genetic risk assessment?

Answer

Women with a family history suggesting increased risk, such as a first degree relative with breast or ovarian cancer, or a known BRCA1 or BRCA2 mutation in the family. This is a Grade B recommendation.

#407 · Preventive Care

Is breast self-exam recommended as a screening method by the USPSTF?

Answer

No. Breast self-exam and routine clinical breast exam are no longer recommended as separate screening maneuvers because they do not add proven benefit beyond mammography.

#408 · Preventive Care

At what age does cervical cancer screening begin, regardless of sexual history?

Answer

Age 21. Screening does not begin earlier even in patients who became sexually active young or have multiple partners, because cervical cancer is rare before this age.

#409 · Preventive Care

What is the cervical screening interval for women aged 21 to 29?

Answer

Cytology alone, meaning a Pap smear, every three years. HPV testing is not used as a primary screening strategy in this younger age group.

#410 · Preventive Care

What are the three acceptable cervical screening strategies for women aged 30 to 65?

Answer

Cytology alone every three years, high risk HPV testing alone every five years, or co-testing with both cytology and HPV testing every five years.

#411 · Preventive Care

When can a woman stop cervical cancer screening?

Answer

After age 65 if she has had adequate prior negative screening and is not otherwise high risk. Women with a total hysterectomy for benign disease also no longer need screening.

#412 · Preventive Care

Which hysterectomy patients still need continued cervical cancer screening?

Answer

Women who had a hysterectomy for cervical cancer or a high grade precancerous lesion. They still need continued screening of the vaginal cuff even after the cervix is removed.

#413 · Preventive Care

At what age does average risk colorectal cancer screening begin?

Answer

Age 45, continuing through age 75. This is a Grade A recommendation, lowered from age 50 due to rising colorectal cancer rates in younger adults.

#414 · Preventive Care

What is the USPSTF recommendation for colorectal screening in adults aged 76 to 85?

Answer

It should be individualized, a Grade C recommendation, weighing health status, prior screening history, and patient preferences. Screening is not recommended after age 85.

#415 · Preventive Care

How often is screening colonoscopy repeated in average risk adults?

Answer

Every ten years. This is one of several accepted colorectal cancer screening intervals, along with FIT annually, sigmoidoscopy every five years, and CT colonography every five years.

#416 · Preventive Care

How often should FIT, fecal immunochemical testing, be repeated?

Answer

Annually. FIT has largely replaced older guaiac based fecal occult blood testing because it has better sensitivity and does not require dietary restriction before testing.

#417 · Preventive Care

When should colorectal screening start in a patient with a first degree relative diagnosed with colorectal cancer at age 45?

Answer

At age 35, which is ten years before the relative's age at diagnosis, since that is earlier than the standard starting age of 40 used for this higher risk group.

#418 · Preventive Care

Who qualifies for annual low dose CT lung cancer screening?

Answer

Adults aged 50 to 80 with a 20 pack year smoking history who currently smoke or quit within the past 15 years. This is a Grade B recommendation.

#419 · Preventive Care

When should lung cancer screening with low dose CT stop?

Answer

Once the person has not smoked for 15 years, or if a health problem develops that substantially limits life expectancy or the ability to have curative lung surgery.

#420 · Preventive Care

What USPSTF grade is PSA screening for men aged 55 to 69?

Answer

Grade C. The decision should be individualized after discussing potential benefits and harms with the patient, rather than ordered automatically without discussion.

#421 · Preventive Care

What USPSTF grade is PSA screening for men 70 and older?

Answer

Grade D. The task force recommends against routine PSA screening in this age group because overdiagnosis and overtreatment risk outweighs the modest potential benefit.

#422 · Preventive Care

At what 10 year ASCVD risk threshold does statin therapy become a Grade B recommendation?

Answer

10 percent or greater, in adults aged 40 to 75 with at least one cardiovascular risk factor such as dyslipidemia, diabetes, hypertension, or smoking.

#423 · Preventive Care

What ASCVD risk range makes statin initiation a Grade C, individualized decision?

Answer

A 10 year risk between 7.5 and 10 percent. Below this range, routine statin initiation for primary prevention is generally not recommended.

#424 · Preventive Care

What inputs does the ASCVD risk calculator use?

Answer

Age, sex, race, total cholesterol, HDL cholesterol, systolic blood pressure, hypertension treatment status, diabetes status, and smoking status, combined to generate a 10 year risk percentage.

#425 · Preventive Care

At what age is universal lipid screening recommended in children?

Answer

Once between ages 9 and 11, and again between ages 17 and 21. This reflects pediatric cardiology guidance outside the USPSTF grading system.

#426 · Preventive Care

At what age should DEXA screening begin for average risk women?

Answer

Age 65, a Grade B recommendation. Women younger than 65 with risk factors that raise fracture risk to that of a 65 year old should be screened earlier.

#427 · Preventive Care

What tool estimates fracture risk to decide if a younger postmenopausal woman needs earlier DEXA screening?

Answer

The FRAX tool, which incorporates factors like low body weight, smoking, alcohol use, family history of hip fracture, and prior fracture history.

#428 · Preventive Care

What is the USPSTF recommendation on osteoporosis screening in men?

Answer

A Grade I statement, meaning current evidence is insufficient. There is no formal task force mandate to screen men, unlike the firm Grade B recommendation for women 65 and older.

#429 · Preventive Care

How often should adults be screened for HIV according to the USPSTF?

Answer

At least once between ages 15 and 65, a Grade A recommendation, with repeat screening for those at higher ongoing risk such as new or multiple sexual partners or injection drug use.

#430 · Preventive Care

When should pregnant women be screened for HIV?

Answer

At the first prenatal visit, and again in the third trimester if they are at high risk or live in a high prevalence area.

#431 · Preventive Care

Who should be screened annually for chlamydia and gonorrhea?

Answer

All sexually active women aged 24 and younger, and older women with risk factors, both Grade B recommendations. Evidence is insufficient for routine screening in asymptomatic men.

#432 · Preventive Care

What is the current USPSTF recommendation on hepatitis C screening in adults?

Answer

Screen once for all adults aged 18 to 79, a Grade B recommendation, reflecting the availability of highly effective direct acting antiviral treatment.

#433 · Preventive Care

Who should receive syphilis screening?

Answer

All pregnant women, and individuals at increased risk, including men who have sex with men and people living with HIV.

#434 · Preventive Care

What tool is used to screen adults for depression, and what USPSTF grade does depression screening carry?

Answer

The PHQ-9, a validated questionnaire. Depression screening in all adults, including pregnant and postpartum women, is a Grade B recommendation.

#435 · Preventive Care

Why should depression screening only be implemented where systems for follow up exist?

Answer

Because screening without accurate diagnosis, effective treatment, and appropriate follow up in place does not improve patient outcomes on its own.

#436 · Preventive Care

What tool is used for anxiety screening in adults younger than 65?

Answer

The GAD-7. Anxiety screening is a Grade B recommendation, added more recently than the long standing depression screening recommendation.

#437 · Preventive Care

What tool screens for unhealthy alcohol use in primary care?

Answer

AUDIT-C, a brief validated tool. Screening is a Grade B recommendation for all adults, followed by brief behavioral counseling for those who screen positive.

#438 · Preventive Care

What are the 5 A's of tobacco use counseling?

Answer

Ask, Advise, Assess, Assist, and Arrange. This is a Grade A recommendation for adults, including pregnant women, and one of the highest yield interventions in primary care.

#439 · Preventive Care

At what BMI is referral to intensive behavioral intervention recommended for obesity?

Answer

A BMI of 30 or greater. Screening for obesity using BMI in all adults is a Grade B recommendation.

#440 · Preventive Care

At what age range does diabetes and prediabetes screening begin for overweight or obese adults?

Answer

Ages 35 to 70, a Grade B recommendation. This was lowered from age 40 in a recent update, a common exam trap for those who learned the older cutoff.

#441 · Preventive Care

What tests can be used to screen for diabetes and prediabetes?

Answer

Fasting glucose, hemoglobin A1c, or an oral glucose tolerance test. All three are acceptable screening options in overweight or obese adults aged 35 to 70.

#442 · Preventive Care

What should happen if a patient screens positive for prediabetes?

Answer

Referral to an effective preventive intervention, such as an intensive lifestyle behavioral counseling program, since this reduces progression to overt type 2 diabetes.

#443 · Preventive Care

How often is influenza vaccination recommended for adults and children?

Answer

Annually, for essentially everyone aged 6 months and older, with rare exceptions for documented severe allergy to a vaccine component.

#444 · Preventive Care

At what age is the recombinant shingles vaccine recommended to start?

Answer

Age 50, given in two doses, regardless of prior shingles history or prior receipt of the older live zoster vaccine.

#445 · Preventive Care

How often should adults receive a Td or Tdap booster?

Answer

Every 10 years. A one time Tdap dose is substituted for one Td booster if the patient has never received Tdap as an adult.

#446 · Preventive Care

Why are live vaccines like MMR and varicella generally avoided in pregnancy and immunocompromise?

Answer

Because a live attenuated vaccine carries theoretical risk of causing disease in a host with reduced immune defenses. Inactivated vaccines remain safe in both situations.

#447 · Preventive Care

When during pregnancy is Tdap recommended?

Answer

During every pregnancy, ideally between 27 and 36 weeks gestation, to maximize passive transfer of pertussis antibodies to the newborn.

#448 · Preventive Care

What influenza vaccine formulation is safe during pregnancy?

Answer

The inactivated formulation, given during any trimester. The live attenuated nasal spray formulation is avoided in pregnancy.

#449 · Preventive Care

When is the RSV vaccine recommended during pregnancy?

Answer

During a defined seasonal window between 32 and 36 weeks gestation, to protect the infant during the first months of life when RSV hospitalization risk is highest.

#450 · Preventive Care

What organization publishes the well child visit and anticipatory guidance schedule used in US pediatric primary care?

Answer

The American Academy of Pediatrics, through the Bright Futures schedule, which structures visits from the newborn period through adolescence.

#451 · Preventive Care

At what well child visits is formal developmental screening with a standardized tool recommended?

Answer

The 9, 18, and 30 month visits, regardless of whether parents have expressed any developmental concerns at that visit.

#452 · Preventive Care

At what ages is autism specific screening recommended?

Answer

18 and 24 months, using a validated autism specific tool, even in children with no parental concerns, since early identification allows earlier intervention.

#453 · Preventive Care

What should prompt further evaluation on a child's growth curve?

Answer

Crossing two major percentile lines in either direction for weight, length or height, or head circumference, since this can signal a growth or nutrition problem.

#454 · Preventive Care

What anticipatory guidance topic is emphasized at the newborn and early infancy visits?

Answer

Safe sleep practices and correct car seat use, since these address the leading preventable causes of infant injury and death at this age.

#455 · Preventive Care

What anticipatory guidance topic is emphasized in the toddler years?

Answer

Poison prevention and water safety, reflecting the toddler's increased mobility and curiosity along with limited danger awareness at this developmental stage.

#456 · Preventive Care

What screening tests are done at the first prenatal visit?

Answer

Blood type and antibody screen, complete blood count, rubella immunity, hepatitis B surface antigen, HIV, syphilis, urine culture, and a Pap smear if due.

#457 · Preventive Care

When is first trimester aneuploidy screening offered in pregnancy?

Answer

Between 10 and 13 weeks gestation, combining nuchal translucency ultrasound with maternal serum markers.

#458 · Preventive Care

When is the fetal anatomy ultrasound typically performed?

Answer

Between 18 and 22 weeks gestation, to evaluate fetal structural development in detail.

#459 · Preventive Care

When is gestational diabetes screening performed?

Answer

Between 24 and 28 weeks gestation, using a glucose challenge test, with a follow up three hour glucose tolerance test if the initial screen is abnormal.

#460 · Preventive Care

When is Group B streptococcus screening performed in pregnancy?

Answer

Between 36 and 37 weeks gestation, close to delivery, since colonization status can change over the course of pregnancy.

#461 · Preventive Care

When is RhoGAM given to Rh negative mothers?

Answer

At 28 weeks gestation, and again after delivery if the infant is confirmed Rh positive, to prevent maternal sensitization in future pregnancies.

#462 · Preventive Care

Why does a Grade C recommendation not carry a mandatory no cost sharing insurance benefit under the ACA?

Answer

Only services graded A or B by the USPSTF are required to be covered by most insurance plans with no patient cost sharing under the Affordable Care Act.

#463 · Preventive Care

What is a common exam trap involving PSA screening in a 72 year old man requesting the test?

Answer

The correct approach is to counsel against routine PSA screening in men 70 and older, a Grade D recommendation, while still respecting patient autonomy after that counseling.

#464 · Preventive Care

What is a common exam trap involving colorectal screening age cutoffs?

Answer

Screening now begins at 45, not the older cutoff of 50, and family history of colorectal cancer or advanced adenoma before age 60 requires starting even earlier.

#465 · Preventive Care

What is a common exam trap involving cervical screening before age 21?

Answer

Screening never begins before age 21 regardless of sexual history or number of partners, and HPV vaccination status does not change this rule.

#466 · Preventive Care

What pack year and quit history thresholds define eligibility for lung cancer screening?

Answer

A 20 pack year smoking history, with current smoking or having quit within the past 15 years, in adults aged 50 to 80.

#467 · Preventive Care

Why is chlamydia and gonorrhea screening not routinely recommended in asymptomatic men?

Answer

The USPSTF states current evidence is insufficient to recommend routine screening in asymptomatic men, unlike the Grade B recommendation for young sexually active women.

#468 · Preventive Care

What distinguishes mammography screening intervals from colorectal screening intervals?

Answer

Mammography is biennial, meaning every two years, while colonoscopy is done every ten years and FIT is done annually, each modality carrying its own distinct interval.

#469 · Preventive Care

What is the difference between statin therapy at 10 percent ASCVD risk versus 8 percent ASCVD risk?

Answer

At 10 percent or greater the recommendation is Grade B, meaning statin therapy should generally be offered. Between 7.5 and 10 percent it is Grade C, an individualized decision.

#470 · Preventive Care

Why is DEXA screening a firm recommendation in women but not in men?

Answer

The USPSTF gives a Grade B recommendation for women 65 and older based on strong evidence, while evidence in men remains insufficient, a Grade I statement.

#471 · Preventive Care

What is the significance of the 5A's framework being Grade A rather than Grade B?

Answer

Tobacco screening and counseling has high certainty of substantial net benefit given the enormous disease burden from smoking, earning the strongest possible recommendation grade.

#472 · Preventive Care

Why must a physician discuss risks and benefits before ordering a Grade C screening test?

Answer

Because Grade C services have only a small net benefit, so the correct clinical approach is shared decision making rather than routine automatic ordering.

#473 · Preventive Care

What ties hepatitis C's shift to a Grade B recommendation for universal adult screening?

Answer

The availability of highly effective direct acting antiviral treatment, which changed the overall risk to benefit calculation in favor of broader screening.

#474 · Preventive Care

Why is knowing exact boundary ages, like 39 versus 40 or 64 versus 65, important for this exam?

Answer

Exam writers frequently place patients exactly at a transition age to test whether the learner has memorized the precise USPSTF cutoff rather than a rough approximation.

#475 · Foundations of Care

What does SnNout mean?

Answer

A highly Sensitive test, when Negative, rules a disease out. Sensitivity is the chance a test is positive when disease is truly present. It does not tell you how likely disease actually is after a positive result.

#476 · Foundations of Care

What does SpPin mean?

Answer

A highly Specific test, when Positive, rules a disease in. Specificity is the chance a test is negative when disease is truly absent. Like sensitivity, it is a fixed property of the test itself.

#477 · Foundations of Care

How does prevalence affect PPV and NPV?

Answer

As disease prevalence falls, positive predictive value falls even if sensitivity and specificity stay the same. This is because false positives make up a larger share of all positive results when true disease is rare. NPV rises as prevalence falls for the same reason.

#478 · Foundations of Care

Do sensitivity and specificity change with disease prevalence?

Answer

No. Sensitivity and specificity are intrinsic properties of the test and stay constant regardless of how common the disease is in the tested population. Only PPV and NPV shift with prevalence.

#479 · Foundations of Care

How is number needed to treat (NNT) calculated?

Answer

NNT equals 1 divided by the absolute risk reduction. Absolute risk reduction is the control group's event rate minus the treatment group's event rate. A lower NNT means a more effective treatment.

#480 · Foundations of Care

How is number needed to harm (NNH) calculated?

Answer

NNH equals 1 divided by the absolute risk increase caused by a treatment. A higher NNH is better because it means the harmful side effect happens less often. NNH and NNT are compared to judge whether a treatment's benefit outweighs its risk.

#481 · Foundations of Care

If a drug cuts a bad outcome from 20 percent to 10 percent, what is the NNT?

Answer

The absolute risk reduction is 20 percent minus 10 percent, which is 10 percent or 0.10. NNT equals 1 divided by 0.10, which is 10. Ten patients need treatment to prevent one additional bad outcome.

#482 · Foundations of Care

What makes a randomized controlled trial the strongest study design for causation?

Answer

Random assignment balances both known and unknown confounding variables between the treatment and control groups. This makes any outcome difference more likely to be caused by the treatment itself. RCTs are limited by cost, time, and exposures that cannot ethically be randomized.

#483 · Foundations of Care

What is the key difference between a cohort study and a case-control study?

Answer

A cohort study groups people by exposure and follows them forward to see who develops the outcome. A case-control study groups people by outcome and looks backward for prior exposures. Case-control studies are efficient for rare diseases but prone to recall bias.

#484 · Foundations of Care

What can and cannot a cross-sectional study tell you?

Answer

A cross-sectional study measures exposure and outcome at a single point in time, so it can estimate prevalence. It cannot establish which came first, the exposure or the outcome, so it cannot prove causation.

#485 · Foundations of Care

What is confounding in a study?

Answer

Confounding happens when a third variable is linked to both the exposure and the outcome, creating a false appearance of association. A classic example is coffee drinking once appearing linked to lung cancer because coffee drinkers were more often smokers. Randomization in an RCT helps control for confounding.

#486 · Foundations of Care

What is recall bias and which study design is most vulnerable to it?

Answer

Recall bias is when study groups remember and report past exposures differently from each other. Case-control studies are especially vulnerable because people with a disease often recall past exposures differently than healthy controls. This can distort the measured association between exposure and outcome.

#487 · Foundations of Care

What does a p-value actually measure?

Answer

A p-value is the probability of seeing the study's result, or something more extreme, by chance alone if there were truly no real effect. A p-value below 0.05 is the conventional cutoff for statistical significance. A p-value does not measure how large or clinically important an effect is.

#488 · Foundations of Care

How do you know if a relative risk or odds ratio is statistically significant from its confidence interval?

Answer

If the 95 percent confidence interval crosses the number 1, the result is not statistically significant, because 1 means no difference between groups. This applies to ratio measures like relative risk and odds ratio, not to difference measures.

#489 · Foundations of Care

What number indicates no effect for an absolute risk difference or mean difference, instead of 1?

Answer

For a difference measure such as an absolute risk difference or a mean difference, the number that indicates no effect is 0, not 1. If the confidence interval for a difference measure crosses 0, the result is not statistically significant.

#490 · Foundations of Care

What are the four required elements of informed consent?

Answer

Disclosure of the diagnosis, treatment, risks, benefits, and alternatives; the patient's understanding of that information; voluntary agreement without coercion; and decision-making capacity. All four elements must be present for consent to be valid.

#491 · Foundations of Care

When can a physician treat without obtaining informed consent first?

Answer

In a genuine emergency where delay to obtain consent would risk life or limb and the patient cannot currently communicate, treatment can proceed under implied consent. This is an exception, not a routine substitute for consent.

#492 · Foundations of Care

Under HIPAA, when can protected health information be shared without specific patient authorization?

Answer

Protected health information can generally be shared without specific authorization for treatment, payment, and health care operations, known as the TPO exceptions. Sharing beyond these purposes, such as with an employer, typically requires the patient's specific authorization.

#493 · Foundations of Care

What is the difference between capacity and competence?

Answer

Capacity is a clinical judgment made by a treating physician about a patient's ability to understand, appreciate, reason through, and communicate a specific medical decision. Competence is a legal determination made by a court about a person's general ability to manage their own affairs. Only a judge can declare someone legally incompetent.

#494 · Foundations of Care

Is decision-making capacity fixed or can it change?

Answer

Capacity is decision-specific and can fluctuate over time. A patient might have capacity for a simple medication decision but lack capacity for a complex surgery decision. The same patient's capacity can also change from morning to evening, for example due to delirium.

#495 · Foundations of Care

What is substituted judgment for a surrogate decision-maker?

Answer

Substituted judgment means the surrogate decides as the patient would have decided, based on the patient's own known values and prior statements. This differs from the surrogate simply choosing what they personally think is best for the patient.

#496 · Foundations of Care

What are social determinants of health?

Answer

The non-medical conditions in which people are born, grow, live, work, and age that shape health outcomes, including income, education, housing, food access, and transportation. These factors often affect population health more than clinical care alone, so primary care should screen for and address them.

#497 · Foundations of Care

What is the key eligibility difference between Medicare and Medicaid?

Answer

Medicare is a federal program primarily based on age, generally 65 and older, plus certain disability and end-stage renal disease cases, and eligibility does not depend on income. Medicaid is a joint federal and state program based primarily on low income and household size, with benefits and rules that vary by state.

#498 · Foundations of Care

What does each part of Medicare cover?

Answer

Part A covers inpatient hospital care, Part B covers outpatient physician services, Part C (Medicare Advantage) is a private-plan alternative bundling Parts A and B, and Part D covers outpatient prescription drugs. Part C often adds extra benefits beyond original Medicare.

#499 · Foundations of Care

What are the four steps of a PDSA quality improvement cycle?

Answer

Plan identifies a problem, sets a measurable goal, and designs a small test of change. Do carries out the test on a limited scale. Study analyzes whether the test produced the intended improvement. Act decides to adopt, adapt, or abandon the change, and the cycle repeats.

#500 · Foundations of Care

What is the difference between a Type I error and a Type II error in a clinical study?

Answer

A Type I error means rejecting a true null hypothesis, concluding there is an effect when there really is not one. A Type II error means failing to reject a false null hypothesis, missing a real effect. Alpha controls the Type I error rate and is usually set at 0.05. Beta controls the Type II error rate, and power equals 1 minus beta.